Exercise J-M

1. For three positive integers p, q, r, $x^{pq^2} = y^{qr} = z^{p^2r}$ and r = pq + 1 such that 3, 3 $\log_y x$, 3$\log_z y$, 7$\log_x z$ are in A.P. with common difference $\frac{1}{2}$. Then r-p-q is equal to : **[JEE(Main)Jan.2023]**

तीन घनात्मक पूर्णांकों p, q, r के लिए $x^{pq^2} = y^{qr} = z^{p^2r}$, r = pq + 1 है तथा 3, 3 $\log_y x$, 3$\log_z y$, 7$\log_x z$ एक A.P. में है, जिसका सार्व अंतर $\frac{1}{2}$ हो तो r-p-q बराबर है : **[JEE(Main)Jan.2023]**

(A) 2 (B) –6 (C) 6 (D) 12

Ans. A

Sol. Here

$$\frac{3\log x}{\log y} - 3 = \frac{1}{2} \Rightarrow \frac{\log x}{\log y} = \frac{7}{6} \quad \text{_____(1)}$$

$$\Rightarrow \frac{3\log y}{\log z} - \frac{3\log y}{\log z} = \frac{1}{2} \Rightarrow \frac{\log y}{\log z} = \frac{4}{3} \quad \text{_____(2)}$$

$$\Rightarrow \frac{7\log z}{\log x} - \frac{3\log y}{\log 2} = \frac{1}{2} \Rightarrow \frac{\log z}{\log x} = \frac{9}{14} \quad \text{_____(3)}$$

Now by $x^{pq^2} = y^{qr} = z^{p^2r}$

$$pq^2 \log x = qr \log y = p^2 r \log z \quad \text{_____(4)}$$

by equation (1), (2) , (3) and (4)

$$\frac{\log x}{\log y} = \frac{r}{pq} = \frac{7}{6} \quad \text{_____(5)}$$

$$\frac{\log y}{\log z} = \frac{p^2}{q} = \frac{4}{3} \quad \text{_____(6)}$$

$$\frac{\log z}{\log x} = \frac{q^2}{pr} = \frac{9}{14} \quad \text{_____(7)}$$

r = pq + 1 (given) _____(8)

By solving equation (5), (6), (7) & (8) we get

r = 7, p = 2 and q = 3

So, r–p–q = 7– 2 – 3 = 2

2. The number of 3 digit numbers, that are divisible by 3 or 4 but not divisible by 48, is : **[JEE(Main)Jan.2023]**

तीन अंकों की संख्याओं, जो 3 या 4 से विभाज्य है परन्तु 48 से विभाज्य नहीं है, की संख्या है : **[JEE(Main)Jan.2023]**

(A) 432 (B) 400 (C) 472 (D) 507

Ans. A

Sol. no. divisible by 3 = 102, 105, ... , 999

= 300

no. divisible by 4 = 100, 104, ... , 996

= 225

now no. divisible by 12

= 75

So numbers divisible by 3 or 4

= 300 + 225 – 75 = 450

now numbers divisible by 48 are = 18

So total numbers = 450 – 18

= 432

3. If $a_n = \frac{-2}{4n^2-16n+15}$, then $a_1 + a_2 + ... + a_{25}$ is equal to : **[JEE(Main)Jan.2023]**

यदि $a_n = \frac{-2}{4n^2-16n+15}$ है, तो $a_1 + a_2 + ... + a_{25}$ बराबर है : **[JEE(Main)Jan.2023]**

(A) $\frac{52}{147}$ (B) $\frac{49}{138}$ (C) $\frac{50}{141}$ (D) $\frac{51}{144}$

Ans. C

Sol. $a_1 + a_2 + ... + a_{25} = \sum_{n=1}^{25} a_n$

$$= \Sigma \frac{-2}{4n^2-16n+15} = \Sigma \frac{-2}{(2n-5)(2n-3)}$$

$$= \sum_{n=1}^{25} \left(\frac{1}{2n-3} - \frac{1}{2n-5} \right)$$

$$= \left[\left(\frac{1}{-1} - \frac{1}{-3} \right) + \left(\frac{1}{1} - \frac{1}{-1} \right) + \left(\frac{1}{3} - \frac{1}{1} \right) + ... \right.$$

$$= \frac{1}{2(25)-3} + \frac{1}{3} = \frac{50}{141}$$

4. Let $a, b, c > 1$, a^3, b^3 and c^3 be in A.P., $\log_a b$, $\log_c a$ and $\log_b c$ be in G.P. If the sum of first 20 terms of an A.P., whose first term is $\frac{a+4b+c}{3}$ and the common difference is $\frac{a-8b+c}{10}$ is -444, then abc is equal to :

[JEE(Main)Jan.2023]

माना $a, b, c > 1$ है, a^3, b^3 व c^3 समान्तर श्रेढी में तथा $\log_a b$, $\log_c a$ व $\log_b c$ गुणोत्तर श्रेढी में है। यदि समान्तर श्रेढी के प्रथम 20 पदों का योग, जिसका प्रथम पद $\frac{a+4b+c}{3}$ है तथा सार्वअंतर $\frac{a-8b+c}{10}$, -444 है। तब abc बराबर है :

[JEE(Main)Jan.2023]

(A) 216 (B) 343 (C) $\frac{125}{8}$ (D) $\frac{343}{8}$

Ans. A

Sol. $2b^3 = a^3 + c^3$

$$\left(\frac{\log a}{\log c}\right)^2 = \frac{\log b}{\log a}.\frac{\log c}{\log b}$$

$\Rightarrow (\log a)3 = (\log c)3$

$\Rightarrow a = c$

$\Rightarrow 2b^3 = 2c^3 \quad \Rightarrow \quad b = c$

$\Rightarrow a = b = c$

$T_1 = 2a, \qquad d = -\frac{3a}{5}$

$$\therefore -444 = 10\left(4a - \frac{57a}{5}\right)$$

$$-444 = -\frac{370a}{5} \quad \Rightarrow \quad a = \frac{444}{74}$$

$\Rightarrow a = 6$

$\Rightarrow abc = 6.6.6 = 216$

5. The number of ways of selecting two numbers a and b, $a \in \{2,4,6,.....,100\}$ and $b \in \{1,3,5,.....,99\}$ such that 2 is the remainder when $a + b$ is divided by 23 is : **[JEE(Main)Jan.2023]**

दो संख्याओं a व b, $a \in \{2,4,6,.....,100\}$ व $b \in \{1,3,5,.....,99\}$ को इस प्रकार चुनने के तरीकों, कि a + b को 23 से विभाजित करने पर शेषफल 2 प्राप्त हो, की संख्या है : **[JEE(Main)Jan.2023]**

(A) 54 (B) 108 (C) 268 (D) 186

Ans. B

Sol. Clearly $\frac{a+b}{23} = \lambda + \frac{2}{23}$, $\lambda \in N$

$\Rightarrow a + b = 23\lambda + 2$

$\Rightarrow a + b$ can be 25, 71, 117, 163

When $a + b = 25 \quad \Rightarrow$ Number of ways = 12

When $a + b = 71 \quad \Rightarrow$ Number of ways = 35

When $a + b = 117 \quad \Rightarrow$ Number of ways = 42

When $a + b = 163 \quad \Rightarrow$ Number of ways = 19

So total number of ways = 108

6. If the sum and product of four positive consecutive terms of a G.P., are 126 and 1296, respectively, then the sum of common ratios of all such GPs is : **[JEE(Main)Jan.2023]**

यदि एक G.P. के चार धनात्मक क्रमागत पदों के योग तथा गुणनफल क्रमशः 126 तथा 1296 है, तो ऐसी सभी G.P. के सार्व अनुपातों का

योग है : **[JEE(Main)Jan.2023]**

(A) 14 (B) $\frac{9}{2}$ (C) 3 (D) 7

Ans. D

Sol. a, ar, ar^2, ar^3 (a, r > 0)

$a^4r^6 = 1296$

$a^2r^3 = 36$

$$a = \frac{6}{r^{3/2}}$$

$a + ar + ar^2 + ar^3 = 126$

$$\frac{1}{r^{3/2}} + \frac{r}{r^{3/2}} + \frac{r^2}{r^{3/2}} + \frac{r^3}{r^{3/2}} = \frac{126}{6} = 21$$

$(r^{-3/2} + r^{3/2}) + (r^{1/2} + r^{-1/2}) = 21$

$r^{1/2} + r^{-1/2} = A$

$r^{-3/2} + r^{3/2} + 3A = A^3$

$A^3 - 3A + A = 21$

$A^3 - 2A = 21$

$A = 3$

$$\sqrt{r} + \frac{1}{\sqrt{r}} = 3$$

$$r + 1 = 3\sqrt{r}$$

$r^2 + 2r + 1 = 9r$

$r^2 - 7r + 1 = 0$

7. Let $a_1, a_2, a_3, \ldots$ be an A.P. If $a_7 = 3$, the product a_1a_4 is minimum and the sum of its first n terms is zero, then $n! - 4a_{n(n+2)}$ is equal to : **[JEE(Main)Jan.2023]**

माना $a_1, a_2, a_3, \ldots$ एक A.P. है। यदि $a_7 = 3$ है, गुणनफल a_1a_4 न्यूनतम है, तथा इसके प्रथम पदों का योग शून्य है, तो $n! - 4a_{n(n+2)}$ बराबर है : **[JEE(Main)Jan.2023]**

(A) $\frac{33}{4}$ (B) 24 (C) 9 (D) $\frac{381}{4}$

Ans. B

Sol. $a_7 = 3 = a + 6d$

$\Rightarrow a = 3 - 6d$

$a_1a_4 = a(a + 3d)$

$= (3 - 6d)(3 - 3d)$

$= 18d^2 - 27d + 9$

$$d_{min} = \frac{27}{2 \times 18} = \frac{3}{4}$$

$$a = \frac{-3}{2} = \frac{-3}{2}$$

$$S_n = \frac{n}{2}[2a + (n-1)d] = 0$$

$$-3 + (n-1)\frac{3}{4} = 0$$

$$\Rightarrow n = 5$$

Now $n! - 4a_{n(n+2)} = 5! - 4a_{35}$

$= 120 - 4(a + 34d)$

$$= 120 - 4\left(\frac{-3}{2} + 34 \times \frac{3}{4}\right)$$

$= 120 + 6 - 102 = 24$

8. The sum to 10 terms of the series $\frac{1}{1+1^2+1^4} + \frac{2}{1+2^2+2^4} + \frac{3}{1+3^2+3^4}$..... is : **[JEE(Main)Jan.2023]**

श्रेणी $\frac{1}{1+1^2+1^4} + \frac{2}{1+2^2+2^4} + \frac{3}{1+3^2+3^4}$..... के दस पदों का योग : **[JEE(Main)Jan.2023]**

(A) $\frac{55}{111}$ (B) $\frac{56}{111}$ (C) $\frac{59}{111}$ (D) $\frac{58}{111}$

Ans. A

Sol. $$t_n = \frac{n}{1+n^2+n^4} = \frac{n}{\left(n^2+n+1\right)\left(n^2-n+1\right)}$$

$$= \frac{1}{2}\left(\frac{1}{\left(n^2-n+1\right)} - \frac{1}{\left(n^2+n+1\right)}\right)$$

$$t_1 = \frac{1}{2}\left(1 - \frac{1}{3}\right)$$

$$t_2 = \frac{1}{2}\left(\frac{1}{3} - \frac{1}{7}\right)$$

$$t_3 = \frac{1}{2}\left(\frac{1}{7} - \frac{1}{13}\right)$$

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$$t_{10} = \frac{1}{2}\left(\frac{1}{91} - \frac{1}{111}\right)$$

$$\therefore S_{10} = t_1 + t_2 + t_3 + ... + t_{10}$$

$$= \frac{1}{2}\left(1 - \frac{1}{111}\right) = \frac{55}{111}$$

9. The 4th term of GP is 500 and its common ratio is $\frac{1}{m}$, $m \in N$. Let S_n denote the sum of the first n terms of this GP.

If $S_6 > S_5 + 1$ and $S_7 < S_6 + \frac{1}{2}$, then the number of possible values of m is _____.

[JEE(Main)Jan.2023]

एक GP का चौथा पद 500 है तथा इसका सार्वअनुपात $\frac{1}{m}$, $m \in N$ है। माना इस GP के प्रथम n पदों का योग S_n है। यदि $S_6 > S_5 + 1$ तथा $S_7 < S_6 + \frac{1}{2}$ हो, तो m के संभव मानों की संख्या है। **[JEE(Main)Jan.2023]**

Ans. 12

Sol. Given 4th term of GP = 500

$$C.R. = \frac{1}{m}$$

$$\frac{a}{m^3} = 500$$

$$S_n = \frac{a\left[\left(\frac{1}{m}\right)^n - 1\right]}{\frac{1}{m} - 1}$$

$$S_n = \frac{a\left(1 - m^n\right)}{m^n \frac{(1-m)}{m}}$$

$$S_n = \frac{a\left(1 - m^n\right)}{m^{n-1}(1-m)}$$

$$S_6 > S_5 + 1$$

$$\frac{a\left(1 - m^6\right)}{m^5(1-m)} > \frac{a\left(1 - m^5\right)}{m^4(1-m)} + 1$$

$$a - am^6 > m^5 - m^6 + am - am^6$$

$$a(1-m) > m^5(1-m)$$

$a > m^5 \quad \ldots(2)$

$S_7 < S_6 + \frac{1}{2}$

$\frac{a\left(1-m^7\right)}{m^6(1-m)} < \frac{a\left(1-m^6\right)}{m^5(1-m)} + \frac{1}{2}$

$a - am^7 < am - am^7 + \frac{1}{2}m^6 - \frac{m^7}{2}$

$a(1-m) < \frac{1}{2}m^6(1-m)$

$a < \frac{m^6}{2} \cdots (3)$

So $m^5 < a < \frac{m^6}{2}$

$m^2 < \frac{a}{m^3} < \frac{m^3}{2}$

$m^2 < 500 < \frac{m^3}{2}$

$m \in [11, 22]$

so, no. of possible value of m is 12

10. If $\frac{1^3 + 2^3 + 3^3 + \ldots.\text{up to n term}}{1.3 + 2.5 + 3.7 + \ldots.\text{up to n term}} = \frac{9}{5}$, then the value of n is _____. **[JEE(Main)Jan.2023]**

यदि $\frac{1^3 + 2^3 + 3^3 + \ldots.\text{ n inkard}}{1.3 + 2.5 + 3.7 + \ldots.\text{ n inkard}} = \frac{9}{5}$, है तो n का मान है। **[JEE(Main)Jan.2023]**

Ans. 5

Sol. $\frac{1^3 + 2^3 + 3^3 + \ldots.\textit{upto n terms}}{1.3 + 2.5 + 3.7 + \ldots.\textit{upto n terms}}$

$\Rightarrow \frac{\frac{n^2(n+1)^2}{4}}{\Sigma n(2n+1)} = \frac{9}{5}$

$$\frac{\frac{n^2(n+1)^2}{4}}{2\cdot\frac{n(n+1)(2n+1)}{6}+\frac{n(n+1)}{2}}=\frac{9}{5}$$

$$\frac{\frac{n(n+1)}{4}\cdot 6}{2(2n+1)+3}=\frac{9}{5}$$

$5n^2 - 19n - 30 = 0$

$n = 5, -\frac{6}{5}$

$n = 5$

11. Let A_1, A_2, A_3, be the three A.P. with the same common difference d and having their first terms as A, A + 1, A + 2, respectively. Let a, b, c be the 7th, 9th, 17th terms of A_1, A_2, A_3, respectively such that $\begin{vmatrix} a & 7 & 1 \\ 2b & 17 & 1 \\ c & 17 & 1 \end{vmatrix} + 70 = 0$. If a = 29, then the sum of first 20 terms of an AP whose first term is c–a–b and common difference is $\frac{d}{12}$, is equal to _____. **[JEE(Main)Jan.2023]**

माना A_1, A_2, A_3 तीन A.P. है, जिनका सार्वअंतर d है तथा जिनके पहले पद क्रमशः A, A + 1, A + 2, है। माना A_1, A_2, A_3, के 7वाँ, 9वाँ, 17वाँ पद क्रमशः a, b, c हैं तथा $\begin{vmatrix} a & 7 & 1 \\ 2b & 17 & 1 \\ c & 17 & 1 \end{vmatrix} + 70 = 0$ है। यदि a = 29 है, तो उस AP जिसका पहला पद c–a–b है तथा सार्वअंतर $\frac{d}{12}$ है, के प्रथम 20 पदों का योग बराबर है। **[JEE(Main)Jan.2023]**

Ans. 495

Sol. A + 6 d = a ____ (1)

A + 1 + 8 d = b _____ (2)

A + 2 + 16 d = c _____ (3)

Now

$$\begin{vmatrix} a & 7 & 1 \\ 2b & 17 & 1 \\ c & 17 & 1 \end{vmatrix} + 70 = 0$$

$\Rightarrow c - 2b = 7$

Using equation (2) & (3) , we get A = –7

Now a = – 7 + 6d = 29 $\Rightarrow$ d = 6

Now first term = c–a–b = 20 and common difference = $\frac{d}{12} = 1/2$ and the sum of first 20 terms of an AP = 495

12. For the two positive numbers a, b, if a, b and $\frac{1}{18}$ are in geometric progression, while $\frac{1}{a}$, 10 and $\frac{1}{b}$ are in an arithmetic progression, then 16a + 12b is equal to _____. **[JEE(Main)Jan.2023]**

दो घनात्मक संख्याओं a, b के लिए, यदि a, b तथा $\frac{1}{18}$ एक गुणोत्तर श्रेढ़ी में है और $\frac{1}{a}$, 10 तथा $\frac{1}{b}$ एक समांतर श्रेढ़ी में है, तो 16a + 12b बराबर है। **[JEE(Main)Jan.2023]**

Ans. 3

Sol. $a, b, \frac{1}{18} \rightarrow GP$

$\frac{a}{18} = b^2$ .....(i)

$\frac{1}{a}, 10, \frac{1}{b} \rightarrow AP$

$\frac{1}{a} + \frac{1}{b} = 20$

$\Rightarrow a + b = 20ab$, from eq. (i) ; we get

$\Rightarrow 18b^2 + b = 360b^3$

$\Rightarrow 360b^2 - 18b - 1 = 0 \quad \{\because b \neq 0\}$

$\Rightarrow b = \frac{18 \pm \sqrt{324 + 1440}}{720}$

$\Rightarrow b = \frac{18 + \sqrt{1764}}{720} \quad \{\because b > 0\}$

$\Rightarrow b = \frac{1}{12}$

$\Rightarrow a = 18 \times \frac{1}{144} = \frac{1}{8}$

Now, $16a + 12b = 16 \times \frac{1}{8} + 12 \times \frac{1}{12} = 3$

13. Let $a_1, a_2, a_3, \ldots\ldots\ldots$ be a GP of increasing positive numbers. If the product of fourth and sixth terms is 9 and the sum of fifth and seventh terms is 24, then $a_1a_9 + a_2a_4a_9 + a_5 + a_7$ is equal to _____ .

[JEE(Main)Jan.2023]

माना $a_1, a_2, a_3, \ldots\ldots$ वर्धमान धनात्मक संख्याओं की एक GP है। यदि चौथे व छटवें पदों का गुणनफल 9 है और पाँचवे व सातवें पदों का योग 24 है, तब $a_1a_9 + a_2a_4a_9 + a_5 + a_7$ बराबर है। **[JEE(Main)Jan.2023]**

Ans. 60

Sol. $\because$ a_1, a_2, a_3 ..... increasing G.P. of positive nos.

$\because$ $a_4a_6 = 9$ and $a_5 + a_7 = 24$

$(ar^3)(ar^5) = 9$

$a^2r^8 = 9$ and $ar^4 + ar^6 = 24$

$\Rightarrow ar^4 = 3$ .....(1) $ar^4(1 + r^2) = 24$ .....(2)

$\therefore$ from (1) and (2) we get $1 + r^2 = 8$

$r^2 = 7 \Rightarrow r = \sqrt{7}$

$\therefore a = \dfrac{3}{49}$

$\therefore$ $a_1.a_9 + a_2\,a_4a_9 + a_5 + a_7$ $= a(ar^8) + (ar)(ar^3)(ar^8) + ar^4 + ar^6$

$= a^2r^8 + a^3r^{12} + ar^4 + ar^6$

$= 9 + 3^3 + 3 + \dfrac{3}{49} \times 7^3$

$= 9 + 27 + 3 + 21$

$= 60$

14. Let $\{a_k\}$ and $\{b_k\}$, $k \in N$, be two G.P.s with common ratios r_1 and r_2 respectively such that $a_1 = b_1 = 4$ and $r_1 < r_2$. Let $c_k = a_k + b_k$, $k \in N$. If $c_2 = 5$ and $c_3 = \dfrac{13}{4}$ then $\sum\limits_{k=1}^{\infty} c_k - \left(12a_6 + 8b_4\right)$ is equal to _____.

[JEE(Main)Jan.2023]

माना $\{a_k\}$ तथा $\{b_k\}$, $k \in N$, दो G.P.s है, जिनके सार्व अनुपात क्रमशः r_1 तथा r_2 है, और $a_1 = b_1 = 4$, $r_1 < r_2$ है। माना $c_k = a_k + b_k$, $k \in N$ है। यदि $c_2 = 5$ तथा $c_3 = \dfrac{13}{4}$ है, तो $\sum\limits_{k=1}^{\infty} c_k - \left(12a_6 + 8b_4\right)$ बराबर है। **[JEE(Main)Jan.2023]**

Ans. 9

Sol. $a = b = 4$

$C_2 = a_2 + b_2 = 5$

$\Rightarrow ar_1 + br_2 = 5$

$\Rightarrow 4r_1 + 4r_2 = 5 \Rightarrow r_1 + r_2 = \dfrac{5}{4}$

$C_3 = a_3 + b_3 = \dfrac{13}{4} \Rightarrow ar_1^2 + br_2^2 = \dfrac{13}{4}$

$\Rightarrow 4r_1^2 + 4r_2^2 = \dfrac{13}{4}$

$\Rightarrow 4\left(\dfrac{5}{4} - r_2\right)^2 + 4r_2^2 = \dfrac{13}{4}$

$$\Rightarrow 16\left(\frac{25}{16}-\frac{5}{2}r_2+r_2^2\right)+16r_2^2=13$$

$$\Rightarrow 25-40r_2+16r_2^2+16r_2^2-13=0$$

$$\Rightarrow 32r_2^2-40r_2+12=0$$

$$\Rightarrow 8r_2^2-10r_2+3=0$$

$$\Rightarrow (4r_2-3)(2r_2-1)=0$$

$$\Rightarrow r_2=\frac{3}{4}\text{or}\frac{1}{2}$$

$$r_1=\frac{1}{2}\text{or}\frac{3}{4}$$

$$r_1<r_2\Rightarrow r_1=\frac{1}{2},r_2=\frac{3}{4}$$

$$\sum_{k=1}^{\infty}C_k=\Sigma a_k+\Sigma b_k=\frac{a}{1-r_1}+\frac{b}{1-r_2}=\frac{4}{1-\frac{1}{2}}+\frac{4}{1-\frac{3}{4}}$$

$$= 8 + 16 = 24$$

$$\sum_{k=1}^{\infty}C_k-12a_6-8b_4=24-12\left(4\left(\frac{1}{2}\right)^5\right)-8.4.\left(\frac{3}{4}\right)^3$$

$$=24-\frac{3}{2}-\frac{27}{2}=9$$

15. Let $a_1 = b_1$ and $a_n = a_{n-1} + (n - 1)$, $b_n = b_{n-1} + a_{n-1}$, $\forall n \geq 2$. If $S=\sum_{n=1}^{10}\frac{b_n}{2^n}$ and $T=\sum_{n=1}^{8}\frac{n}{2^{n-1}}$, then 2^7 (2S - T) is equal to _____. **[JEE(Main)Jan.2023]**

माना $a_1 = b_1$ तथा $a_n = a_{n-1} + (n - 1)$, $b_n = b_{n-1} + a_{n-1}$, $\forall n \geq 2$ है। यदि $S=\sum_{n=1}^{10}\frac{b_n}{2^n}$ तथा $T=\sum_{n=1}^{8}\frac{n}{2^{n-1}}$ हैं, तो 2^7 (2S - T) बराबर है। **[JEE(Main)Jan.2023]**

Ans. 461

Sol. $$S=\frac{b_1}{2}+\frac{b_2}{2^2}+\frac{b_3}{2^3}+.....+\frac{b_{10}}{2^{10}} \qquad (A)$$

$$\frac{S}{2}=\frac{b_1}{2^2}+\frac{b_2}{2^3}+\frac{b_3}{2^4}+.....+\frac{b_{10}}{2^{11}} \qquad (B)$$

$$(A)-(2)\Rightarrow\frac{S}{2}=\frac{b_1}{2}+\frac{a_1}{2^2}+\frac{a_2}{2^3}+.....+\frac{a_9}{2^{10}}-\frac{b_{10}}{2^{11}}$$

$$\Rightarrow S=b_1+\frac{a_1}{2}+\frac{a_2}{2^2}+.....+\frac{a_9}{2^9}-\frac{b_{10}}{2^{10}} \qquad (C)$$

$$\Rightarrow \frac{S}{2} = \frac{b_1}{2} + \frac{a_1}{2^2} + \ldots\ldots\ldots + \frac{a_9}{2^{10}} - \frac{b_{10}}{2^{11}} \qquad \text{(D)}$$

$$\text{(B)} - \text{(4)} \Rightarrow \frac{S}{2} = \frac{b_1}{2} + \frac{a_1}{2} + \frac{1}{2^2} + \frac{2}{2^3} \ldots\ldots\ldots + \frac{8}{2^9} - \frac{a_9}{2^{10}} - \frac{b_{10}}{2^{11}}$$

$$\Rightarrow \frac{S}{2} = \frac{1}{2} + \frac{1}{2} + \frac{1}{2^2}\left(\frac{1}{1} + \frac{2}{2} + \ldots\ldots\ldots + \frac{8}{2^7}\right) - \frac{a_9}{2^{10}} - \frac{b_{10}}{2^{11}}$$

$$\Rightarrow \frac{S}{2} = 1 + \frac{1}{4}(T) - \frac{a_9}{2^{10}} - \frac{b_{10}}{2^{11}}$$

$$\Rightarrow 2S = 4 + T - \frac{a_9}{2^8} - \frac{b_{10}}{2^9}$$

$$\Rightarrow 2S - T = 4 - \frac{a_9}{2^8} - \frac{b_{10}}{2^9}$$

$$\Rightarrow 2^7(2S - T) = 2^9 - \frac{37}{2} - \frac{130}{2^2}$$

$= 461$

16. The total number of 4-digit numbers whose greatest common divisor with 54 is 2, is _____.

[JEE(Main)Jan.2023]

4-अंकों की संख्याओं, जिनका 54 के साथ महत्तम उभयनिष्ठ भाजक 2 है, की कुल संख्या है। **[JEE(Main)Jan.2023]**

Ans. 3000

Sol. Four digit numbers which are divisible by $2 \Rightarrow 1000$ to $9998 \Rightarrow 4500$

Four digit numbers divisible by 6

$\Rightarrow 1002$ to $9996 \Rightarrow 1500$

Four digit integer divisible by 2 but not divisible by $3 \Rightarrow 4500 - 1500 = 3000$

17. The 8th common term of the series

$S_1 = 3 + 7 + 11 + 15 + 19 + \ldots\ldots\ldots,$

$S_2 = 1 + 6 + 11 + 16 + 21 + \ldots\ldots\ldots$ is _____. **[JEE(Main)Jan.2023]**

श्रेणियों

$S_1 = 3 + 7 + 11 + 15 + 19 + \ldots\ldots\ldots,$

$S_2 = 1 + 6 + 11 + 16 + 21 + \ldots\ldots\ldots$

का 8वाँ उभयनिष्ठ पद है। **[JEE(Main)Jan.2023]**

Ans. 151

Sol. $T_8 = 11 + (8 - 1) \times 20$

$= 11 + 140 = 151$

18. Let $a_1, a_2, ..., a_n$ be in A.P. If $a_5 = 2a_7$ and $a_{11} = 18$, then

$$12\left(\frac{1}{\sqrt{a_{10}}+\sqrt{a_{11}}}+\frac{1}{\sqrt{a_{11}}+\sqrt{a_{12}}}+....+\frac{1}{\sqrt{a_{17}}+\sqrt{a_{18}}}\right)$$ is equal to _____.

[JEE(Main)Jan.2023]

माना $a_1, a_2, ..., a_n$ A.P में है। यदि $a_5 = 2a_7$ तथा $a_{11} = 18$ है, तो

$$12\left(\frac{1}{\sqrt{a_{10}}+\sqrt{a_{11}}}+\frac{1}{\sqrt{a_{11}}+\sqrt{a_{12}}}+....+\frac{1}{\sqrt{a_{17}}+\sqrt{a_{18}}}\right)$$ बराबर है। **[JEE(Main)Jan.2023]**

Ans. 8

Sol. Given $a + 4d = 2(a + 6d)$

$a + 8d = 0$ ..... (1)

and $a_{11} = 18 \Rightarrow a + 10d = 18$ ..... (2)

From (1) and (2)

$-8d + 10d = 18 \Rightarrow d = 9$ and $a = -72$

$a_{10} = a + 9d = -72 + 81 = 9$

$a_{18} = a + 17d = -72 + 153 = 81$

Now

$$12\left[\frac{1}{\sqrt{a_{10}}+\sqrt{a_{11}}}+\frac{1}{\sqrt{a_{11}}+\sqrt{a_{12}}}+.....+\frac{1}{\sqrt{a_{17}}+\sqrt{a_{18}}}\right]$$

$$12\left[\frac{\sqrt{a_{10}}-\sqrt{a_{11}}}{-d}+\frac{\sqrt{a_{11}}-\sqrt{a_{12}}}{-d}+.....+\frac{\sqrt{a_{17}}-\sqrt{a_{18}}}{-d}\right]$$

$$12\left[\frac{\sqrt{a_{10}}-\sqrt{a_{18}}}{-d}\right]=12\left[\frac{3-9}{-9}\right]=12\times\frac{6}{9}=8$$

19. Number of 4-digit numbers that are less than or equal to 2800 and either divisible by 3 or by 11, is equal to _____. **[JEE(Main)Jan.2023]**

4-अंकों की संख्याओं, जो 2800 से कम या इसके बराबर है तथा 3 या 11 से विभाज्य है, की संख्या है।

[JEE(Main)Jan.2023]

Ans. 710

Sol. 1000 – 2799

Divisible by 3

$1002 + (n - 1)3 = 2799$

n = 600

Divisbible by 11

$1-2799 \to \left[\frac{2799}{11}\right] = [254] = 254$

$1-999 = \left[\frac{999}{11}\right] = 90$

$1000 - 2799 = 254 - 90 = 164$

Divisible by 33

$1-2799 \to \left[\frac{2799}{33}\right] = 84$

$1-999 \to \left[\frac{999}{33}\right] = 30$

$1000 - 2799 \to 54$

$\therefore n(3) + n(11) - n(33)$

$600 + 164 - 54 = 710$

20. The sum $1^2 - 2\cdot3^2 + 3\cdot5^2 - 4\cdot7^2 + 5\cdot9^2 - ... + 15\cdot29^2$ is _____ . **[JEE(Main)Jan.2023]**

$1^2 - 2\cdot3^2 + 3\cdot5^2 - 4\cdot7^2 + 5\cdot9^2 - ... + 15\cdot29^2$ बराबर है। **[JEE(Main)Jan.2023]**

Ans. 6952

Sol. Separating odd placed and even placed terms we get

$$S = \left(1.1^2 + 3.5^2 + ...15.(29)^2\right) - \left(2.3^2 + 4.7^2 + ... + 14.(27)^2\right.$$

$$S = \sum_{n=1}^{8}(2n-1)(4n-3)^2 - \sum_{n=1}^{7}(2n)(4n-1)^2$$

Applying summation formula we get = 29856 – 22904 = 6952

21. Let $a_1, = 8, a_2, a_3, ...a_n$, be an A.P. If the sum of its first four terms is 50 and the sum of its last four terms is 170, then the product of its middle two terms is _____. **[JEE(Main)Jan.2023]**

माना $a_1, = 8, a_2, a_3, ...a_n$, एक A.P. है। यदि इसके प्रथम चार पदों का योग 50 है तथा इसके अन्तिम चार पदों का योग 170 है, तब इसके मध्य दो पदों का गुणनफल है। **[JEE(Main)Jan.2023]**

Ans. 754

Sol. $S_4 = 50$

$2(16 + 3d) = 50$

$d = 3$

$4a + d(4n - 10) = 170$

$32 + 3(4n - 10) = 170$

$4n - 10 = 46$

$n = 14$

Middle terms are T_7, T_8

$T_7T_8 = (8 + 6 \times 3)(8 + 7 \times 3) = 26 \times 29$

$= 754$

22. The sum of the common terms of the following three arithmetic progressions.

3,7,11,15,.....,399,

2,5,8,11,.....,359 and

2,7,12,17,.....,197,

is equal to _____ . **[JEE(Main)Jan.2023]**

तीन समांतर श्रेणियों

3,7,11,15,.....,399,

2,5,8,11,.....,359 तथा

2,7,12,17,.....,197,

के उभयनिष्ठ पदों का योग है। **[JEE(Main)Jan.2023]**

Ans. 321

Sol. $S_1 = \{2,5,8,11,14, _____, 359\}$ S_2

$S_2 = \{3,7,11,15, _____, 239\}$ S_1

$S_3 = \{7,12,17,22, _____, 197\}$ S_3

common AP

$S = \{47,107,167\}$

Hence sum of common AP = 321

23. The sum of the first 20 terms of the series $5 + 11 + 19 + 29 + 41 +$_____. **[JEE(Main)April,2023]**

श्रेणी $5 + 11 + 19 + 29 + 41 +$ ..... के प्रथम 20 पदों का योग है : **[JEE(Main)April,2023]**

(A) 3520 (B) 3420 (C) 3250 (D) 3450

Ans. A

Sol. $S_{20} = 5 + 11 + 19 + 29 + \ldots\ldots$

Let $T_r = ar^2 + br + c$

$T_1 = a + b + c = 5$

$T_2 = 4a + 2b + c = 11$

$T_3 = 9a + 3b + c = 19$

$a = 1, b = 3, c = 1$

Hence $S_{20} = \sum_{r=1}^{20} r^2 + 3\sum_{r=1}^{20} r + \sum_{r=1}^{20} 1 = 3520$

24. If gcd (m, n) = 1 and $1^2-2^2+3^2-4^2+....+(2021)^2-(2022)^2+(2023)^2=1012m^2n$ then m^2-n^2 is equal to:

[JEE(Main)April,2023]

यदि gcd (m, n) = 1 है तथा $1^2-2^2+3^2-4^2+....+(2021)^2-(2022)^2+(2023)^2=1012m^2n$ है, तो m^2-n^2 बराबर है:

[JEE(Main)April,2023]

(A) 220 (B) 200 (C) 240 (D) 180

Ans. C

Sol. $1^2-2^2+3^2-4^2+\ldots(2021)^2-(2022)^2+(2023)^2=1012$

$m^2n=(1-2)(1+2)+(3-4)(3+4)+\ldots+(2021-2022)(2021+2022)+(2023)^2$

$=(-1)(1+2+3+4+....+2022)+(2023)^2$

$=(-1)\cdot\frac{(2022)(2023)}{2}+(2023)^2$

$=2023(2023-1011)=2023\times1012$

$m^2n=2023=17^2\cdot7$

$m=17,n=7$

$m^2-n^2=17^2-7^2=240$

25. Let $S_K=\frac{1+2+\ldots+K}{K}$ and $\sum_{j=1}^{n}S_j^2=\frac{n}{A}\left(Bn^2+Cn+D\right)$, where $A,B,C,D\in N$ and A has least value. Then :

[JEE(Main)April,2023]

(A) A + B is divisible by D (B) A + B + C + D is divisible by 5

(C) A + B = 5(D – C) (D) A + C + D is not divisible by B

माना $S_K=\frac{1+2+\ldots+K}{K}$ तथा $\sum_{j=1}^{n}S_j^2=\frac{n}{A}\left(Bn^2+Cn+D\right)$ है, जहाँ $A,B,C,D\in N+$ है तथा A का मान न्यूनतम है। तो :

[JEE(Main)April,2023]

(A) A + B, D से विभाज्य है (B) A + B + C + D, 5 से विभाज्य है

(C) A + B = 5(D – C) (D) A + C + D, B से विभाज्य नहीं है

Ans. A

Sol. $S_K=\frac{K\cdot(K+1)}{2K}=\frac{K+1}{2}$

$$\sum_{j=1}^{n}\left(S_j\right)^2=\sum_{j=1}^{n}\frac{(j+1)^2}{4}=\frac{1}{4}\left(2^2+3^2+4^2\ldots(n+1)^2\right)$$

$$\Rightarrow\frac{1}{4}\left(\frac{(n+1)(n+2)(2n+3)}{6}-1\right)$$

$$=\frac{1}{4}\left(\frac{\left(n^2+3n+2\right)(2n+3)-6}{6}\right)$$

$$=\frac{1}{4}\left(\frac{2n^3+6n^2+4n+3n^2+9n+6-6}{6}\right)$$

$$=\frac{1}{4}\left(\frac{2n^3+9n^2+13n}{6}\right)$$

$$=\frac{n}{24}\left(2n^2+9n+13\right)$$

$A=24, B=2, C=9, D=13$

$$\frac{A+B}{D}=\frac{26}{13}=2$$

26. Let a_n be the nth term of the series $5 + 8 + 14 + 23 + 35 + 50 +$..... and $S_n=\sum_{k=1}^{n}a_k$. Then $S_{30} - a_{40}$ is equal to:

[JEE(Main)April,2023]

माना श्रेणी $5 + 8 + 14 + 23 + 35 + 50 +$..... का nवाँ पद a_n है तथा $S_n=\sum_{k=1}^{n}a_k$ तो $S_{30} - a_{40}$ बराबर है :

[JEE(Main)April,2023]

(A) 11260 (B) 11310 (C) 11280 (D) 11290

Ans. D

Sol. $S_n=5+8+14+23+35+\ldots\ldots+a_{n-1}+a_n$

$S_n=\quad 5+8+14+23\ldots\ldots\ldots\ldots\ldots\ldots+a_{n-1}+a_n$

$a_n=5+3+6+9+12+\ldots\ldots\ldots$

$$a_n=5+\frac{n-1}{2}[6+(n-2)3]$$

$$a_n=5+\left(\frac{n-1}{2}\right)(3n)=\frac{3}{2}n(n-1)+5$$

$$a_n=\frac{3n^2}{2}-\frac{3n}{2}+5$$

$$a_{40} = 3\times\frac{40^2}{2} - 3\times\frac{40}{2} + 5$$

$$= 2400 - 60 + 5 = 2345$$

$$S_{30} = \frac{3}{2}\times\frac{30\times31\times61}{6} - \frac{3}{2}\frac{30\times31}{2} + 5\times30$$

$$= 13635$$

$$S_{30} - a_{40} = 11290$$

27. Let the first term a and the common ratio r of a geometric progression be positive integers. If the sum of squares of its first three terms is 33033 , then the sum of these three terms is equal to : **[JEE(Main)April,2023]**

माना एक गुणोतर श्रेढ़ी के प्रथम पद a तथा सार्व अनुपात r धनातम पूर्णांक है। यदि इसके प्रथम तीन पदों के वर्गों का योग 33033 है, तो इन तीन पदों का योग है : **[JEE(Main)April,2023]**

(A) 231 (B) 220 (C) 210 (D) 241

Ans. A

Sol. $a^2 + a^2r^2 + a^2r^4 = 33033$

$$a^2\left(1 + r^2 + r^4\right) = 33033$$

$$a^2\left(1 + r^2 + r^4\right) = 3\times7\times(11)^2\times13$$

$$\Rightarrow \quad a^2 = (11)^2$$

$a = 11$

$$\Rightarrow \quad 1 + r^2 + r^4 = 273$$

$$r^4 + r^2 - 272 = 0$$

$$\Rightarrow \quad r^2 = 16$$

$r = 4$

$a = 11$

$ar = 44$

$ar^2 = 176$

$a + ar + ar^2 = 231$

28. If $S_n = 4 + 11 + 21 + 34 + 50 + \ldots$ to n term, then $\frac{1}{60}\left(S_{29} - S_9\right)$ is equal to : **[JEE(Main)April,2023]**

यदि $S_n = 4 + 11 + 21 + 34 + 50 + \ldots$ n पदों तक है, तो $\frac{1}{60}\left(S_{29} - S_9\right)$ बराबर है : **[JEE(Main)April,2023]**

(A) 226 (B) 227 (C) 220 (D) 223

Ans. D

Sol. $S = 4 + 11 + 21 + 34 + 50 + \ldots + T_n$

$$S = 4 + 11 + 21 + 34 + \ldots + T_{n-1} + T_n$$

$$T_n = 4 + 7 + 10 + 13 + \ldots$$

$$T_n = 4 + \frac{1}{2}\left(3n^2 + 5n - 8\right)$$

$$\sum T_n = S_n = \frac{3}{2}\sum n^2 + \frac{5}{2}\sum n$$

$$= \frac{3}{2}\frac{n(n+1)(2n+1)}{6} + \frac{5}{2}\frac{n(n+1)}{2}$$

$$\frac{S_{29} - S_9}{60} = 223$$

29. Let $x_1, x_2, ..., x_{100}$ be in an arithmetic progression, with $x_1 = 2$ and their mean equal to 200. If $y_i = i\left(x_i - i\right), 1 \le i \le 100$, then the mean of $y_1, y_2, ..., y_{100}$ is : **[JEE(Main)April,2023]**

माना $x_1, x_2, ..., x_{100}$ एक समांतर श्रेणी में हैं, जिनका माध्य 200 है तथा $x_1 = 2$ है। यदि $y_i = i\left(x_i - i\right), 1 \le i \le 100$ हैं, तो $y_1, y_2, ..., y_{100}$ का माध्य है : **[JEE(Main)April,2023]**

(A) 10051.50 (B) 10049.50 (C) 10101.50 (D) 10100

Ans. B

Sol. $2 + (2 + d) + ... + (2 + 99d) = 20000$

$$\Rightarrow 200 + \frac{99}{2} \times 100d = 20000$$

$$\Rightarrow \quad 99d = 396$$

$$d = 4$$

$$y_1 + y_2 + ... + y_{100} = \sum_{i=1}^{100} i(2 + (i-1)4 - i) = \sum_{i=1}^{100} \left(3i^2 - 2i\right) = \frac{3.100.101.201}{6} - 100.101$$

$$= 10100\left(\frac{201}{2} - 1\right) = (10100).\left(\frac{199}{2}\right)$$

$$\text{So mean} = \frac{(10100)(99.5)}{100} = 10049.5$$

30. Let a, b, c and d be positive real numbers such that a + b + c + d = 11. If the maximum value of $a^5b^3c^2d$ is 3750β, then the value of β is : **[JEE(Main)April,2023]**

माना a, b, c तथा d धनात्मक वास्तविक संख्याएँ हैं तथा a + b + c + d = 11 है। यदि $a^5b^3c^2d$ का उच्चतम मान 3750β है, तो β का मान है : **[JEE(Main)April,2023]**

(A) 108 (B) 110 (C) 90 (D) 55

Ans. C

Sol. Given a + b + c + d = 11 (a, b, c, d > 0}

$(a^5b^3c^2d)_{max.} = ?$

Let assume Numbers –

$\frac{a}{5},\frac{a}{5},\frac{a}{5},\frac{a}{5},\frac{a}{5},\frac{b}{3},\frac{b}{3},\frac{b}{3},\frac{c}{2},\frac{c}{2},$

We know A.M. ≥ G.M.

$$\frac{\frac{a}{5}+\frac{a}{5}+\frac{a}{5}+\frac{a}{5}+\frac{a}{5}+\frac{b}{3}+\frac{b}{3}+\frac{b}{3}+\frac{c}{2}+\frac{c}{2}+d}{11} \geq \left(\frac{a^5b^3c^2d}{5^5\cdot 3^3\cdot 2^2\cdot 1}\right)^{\frac{1}{11}}$$

$$\frac{11}{11} \geq \left(\frac{a^5b^3c^2d}{5^5\cdot 3^3\cdot 2^2\cdot 1}\right)^{\frac{1}{11}}$$

$a^5\cdot b^3\cdot c^2\cdot d \leq 5^5\cdot 3^3\cdot 2^2$

$\max\left(a^5b^3c^2d\right) = 5^5\cdot 3^3\cdot 2^2 = 337500$

$= 90\times 3750 = \beta\times 3750$

$\beta = 90$

31. Let $<a_n>$ be a sequence such that $a_1+a_2+\ldots+a_n=\frac{n^2+3n}{(n+1)(n+2)}$. If $28\sum_{k=1}^{10}\frac{1}{a_k}=p_1p_2p_3\ldots p_m$, where $p_1, p_2, \ldots\ldots p_m$ are the first m prime numbers, then m equal to : **[JEE(Main)April,2023]**

माना $<a_n>$ एक अनुक्रम है जिसके लिए $a_1+a_2+\ldots+a_n=\frac{n^2+3n}{(n+1)(n+2)}$ है। यदि $28\sum_{k=1}^{10}\frac{1}{a_k}=p_1p_2p_3\ldots p_m$ है, जहाँ $p_1, p_2, \ldots\ldots p_m$ प्रथम m अभाज्य संख्या है तो m बराबर है : **[JEE(Main)April,2023]**

(A) 5 (B) 7 (C) 6 (D) 8

Ans. C

Sol. $a_n=\frac{n^2+3n}{(n+1)(n+2)}-\frac{(n-1)^2+3(n-1)}{n(n+1)}$

$$a_n=\frac{n^3+3n^2-(n+2)\left(n^2+n-2\right)}{n(n+1)(n+2)}$$

$$a_n=\frac{4}{n(n+1)(n+2)}$$

$$\sum_{k=1}^{10}\frac{1}{a_k}=\frac{1}{4}\sum_{k=1}^{10}(k(k+1)(k+2))$$

$$\sum_{k=1}^{10}\frac{1}{a_k}=\frac{1}{16}\sum_{k=1}^{10}(k(k+1)(k+2)(k+3)-(k-1)k(k+1)(k+2))$$

$$\sum_{k=1}^{10}\frac{1}{a_k}=\frac{1}{16}(10\cdot 11\cdot 12\cdot 13)$$

$$28\sum_{k=1}^{10}\frac{1}{a_k}=\frac{28}{16}\cdot 2\cdot 5\cdot 11\cdot 3\cdot 4\cdot 13 \quad \Rightarrow m=6$$

32. Let $s_1, s_2, s_3, \ldots\ldots\ldots s_{10}$ respectively be the sum to 12 terms of 10 A.P. s whose first terms are 1, 2, 3, ……..,10 and the common differences are 1, 3, 5, …….19 respectively. Then $\sum_{i=1}^{10} s_i$ is equal to :

[JEE(Main)April,2023]

माना 10 A.P., जिनके प्रथम पद 1, 2, 3, ……..,10 तथा सार्व अंतर क्रमशः 1, 3, 5, ……., 19 हैं, के 12 पदों का योग क्रमशः $s_1, s_2, s_3, \ldots\ldots\ldots s_{10}$ है। तो $\sum_{i=1}^{10} s_i$ बराबर है : **[JEE(Main)April,2023]**

(A) 7380 (B) 7360 (C) 7260 (D) 7220

Ans. C

Sol. $1 \le r \le 10$

$$S_r=\frac{12}{2}\left[2r+(12-1)\cdot(2r-1)\right]=6\left(2r+22r-11\right)$$

$= 6\,(24r - 11)$

$$\sum_{r=1}^{10} S_r = 6\sum_{r=1}^{10}(24r-11)=6\times\frac{10}{2}\left[24-11+240-11\right]$$

$= 30\,[242] = 7260$

33. Let $a_1, a_2, a_3, \ldots\ldots$ be a G. P. of increasing positive numbers. Let the sum of its 6th and 8th terms be 2 and the product of its 3rd and 5th terms be $\frac{1}{9}$. Then $6(a_2 + a_4)(a_4 + a_6)$ is equal to : **[JEE(Main)April,2023]**

माना $a_1, a_2, a_3, \ldots\ldots$ वर्धमान धनात्मक संख्याओं की एक G. P. है। माना इसके छठे और आठवें पदों का योग 2 है तथा इसके तीसरे और पाँचवें पदों का गुणनफल $\frac{1}{9}$ है, तो $6(a_2 + a_4)(a_4 + a_6)$ बराबर है : **[JEE(Main)April,2023]**

(A) 3 (B) 2 (C) $3\sqrt{3}$ (D) $2\sqrt{2}$

Ans. A

Sol. $a_6 + a_8 = 2$

$\Rightarrow ar^5 + ar^7 = 2$ ...(i)

$$a_3\cdot a_5=\frac{1}{9}\Rightarrow a^2\cdot r^2\cdot r^4=\frac{1}{9}$$

$$\Rightarrow ar^3=\frac{1}{3}$$

$$\frac{r^2}{3}+\frac{r^4}{3}=2$$

$\Rightarrow r^4 + r^2 = 6$

$\Rightarrow (r^2 + 3)(r^2 - 2) = 0$

$\Rightarrow r^2 = 2$

$\therefore ar \cdot 2 = \frac{1}{3} \Rightarrow ar = \frac{1}{6}$

Now, $6(a_2 + a_4)(a_4 + a_6)$

$= 6(ar + ar^3)(ar^3 + ar^5)$

$= 6\left(\frac{1}{6} + \frac{1}{3}\right)\left(\frac{1}{3} + \frac{2}{3}\right)$

$= 6 \cdot \frac{1}{2} \cdot 1 = 3$

34. Let A_1 and A_2 be two arithmetic means and G_1, G_2, G_3 be three geometric means of two distinct positive numbers. Then $G_1^4 + G_2^4 + G_3^4 + G_1^2G_3^2$ is equal to : **[JEE(Main)April,2023]**

यदि दो धनात्मक विभिन्न संख्याओं के मध्य दो समान्तर माध्य क्रमशः A_1, A_2 तथा तीन गुणोत्तर माध्य G_1, G_2, G_3 है, तो $G_1^4 + G_2^4 + G_3^4 + G_1^2G_3^2$ का मान बराबर होगा : **[JEE(Main)April,2023]**

(A) $(A_1 + A_2)G_1^2G_3^2$ (B) $2(A_1 + A_2)G_1G_3$ (C) $(A_1 + A_2)^2 G_1G_3$ (D) $2(A_1 + A_2)G_1^2G_3^2$

Ans. C

Sol. Let the two numbers are a, b.

$A_1 = a + \frac{b-a}{3} = \frac{2a+b}{3}$

$A_2 = a + \frac{b-a}{3} \cdot 2 = \frac{a+2b}{3}$

$G_1 = a\left(\frac{b}{a}\right)^{\frac{1}{4}}$

$G_2 = a\left(\frac{b}{a}\right)^{\frac{2}{4}}$

$G_3 = a\left(\frac{b}{a}\right)^{\frac{3}{4}}$

$(G_1)^4 + (G_2)^4 + (G_3)^4 + (G_1)^2 \cdot (G_3)^2$

$\Rightarrow a^4 \cdot \frac{b}{a} + a^4 \cdot \frac{b^2}{a^2} + a^4 \cdot \frac{b^3}{a^3} + a^4 \cdot \frac{b^2}{a^2}$

$= ba^3 + b^2a^2 + b^3a + a^2b^2$

$= ab(a^2 + b^2 + 2ab) = ab(a + b)^2$

$(A_1 + A_2)^2 \cdot G_1G_3 = (a + b)^2 \cdot ab$

35. If $(20)^{19}+2(21)(20)^{18}+3(21)^2(20)^{17}+\ldots+20(21)^{19}=k(20)^{19}$, then k is equal to _____.

[JEE(Main)April,2023]

यदि $(20)^{19}+2(21)(20)^{18}+3(21)^2(20)^{17}+\ldots+20(21)^{19}=k(20)^{19}$ है, तो k बराबर है _____

[JEE(Main)April,2023]

Ans. 400

Sol. $k=1+2\left(\frac{21}{20}\right)+3\left(\frac{21}{20}\right)^2+\ldots\ldots+20\left(\frac{21}{20}\right)^{19}$

$k=1+2r+3r^2+4r^3+\ldots\ldots 20r^{19}$ where $r=\frac{21}{20}$

$kr=r+2r^2+3r^3+\ldots\ldots+20r^{20}\ldots\ldots\ldots(2)$

$by(1)-(2)$

$(1-r)k=1+r+r^2+\ldots\ldots r^{19}-20r^{20}$

$-\frac{1}{20}k=1\frac{\left(r^{20}-1\right)}{r-1}-20r^{20}$

$-\frac{1}{20}k=\frac{\left(\frac{21}{20}\right)^{20}-1}{\left(\frac{21}{20}-1\right)}-20\left(\frac{21}{20}\right)^{20}$

$-\frac{1}{20}k=20\left(\frac{21}{20}\right)^{20}-20-20\left(\frac{21}{20}\right)^{20}$

$k=400$

36. Let $0<z<y<x$ be three real numbers such that $\frac{1}{x},\frac{1}{y},\frac{1}{z}$ are in an arithmetic progression and $x,\sqrt{2}y,z$ are in a geometric progression. If $xy+yz+zx=\frac{3}{\sqrt{2}}xyz$, then $3(x+y+z)^2$ is equal to _____.

[JEE(Main)April,2023]

माना तीन वास्तविक संख्याएँ $0<z<y<x$ इस प्रकार हैं कि $\frac{1}{x},\frac{1}{y},\frac{1}{z}$ एक समांतर श्रेढ़ी में हैं तथा $x,\sqrt{2}y,z$ एक गुोत्तर श्रेढ़ी में हैं। यदि $xy+yz+zx=\frac{3}{\sqrt{2}}xyz$ है, तो $3(x+y+z)^2$ बराबर है _____ **[JEE(Main)April,2023]**

Ans. 150

Sol. As given, $\frac{2}{y}=\frac{1}{x}+\frac{1}{z}$ ....(i)

$2y^2=xz$ ....(ii)

and $\frac{1}{x}+\frac{1}{y}+\frac{1}{z}=\frac{3}{\sqrt{2}}$ ....(iii)

From (i) and (iii) $y=\sqrt{2}$ ....(iv)

Now from (ii) $xz=4$ ....(v)

Now using (ii), (iv) and (v)

$x+z=4\sqrt{2}$

Required, $3(x+y+z)^2=3(\sqrt{2}+4\sqrt{2})^2$

$=150$

37. The sum of all those terms, of the arithmetic progression 3, 8, 13, ..., 373, which are not divisible by 3, is equal to _____. **[JEE(Main)April,2023]**

समांतर श्रेढ़ी 3, 8, 13, ..., 373 के उन सभी पदों, जो 3 से विभाज्य नहीं है, का योग बराबर है _____

[JEE(Main)April,2023]

Ans. 9525

Sol. $T_n=3+5(n-1)=5n-2$

$\Rightarrow T_1, T_4, T_7 \ldots$ are divisible by 3

i.e. 3, 18, 33, 48, ..., 363

Sum of numbers divisible by 3.

$=\frac{25}{2}(3+363)=(25)(183)$

Sum of all numbers in A.P.

$=\frac{75}{2}(3+373)=(75)(188)$

Required sum = 9525

38. Suppose a_1, a_2, 2, a_3, a_4 be in an arithmetico-geometric progression. If the common ratio of the corresponding geometric progression is 2 and the sum of all 5 terms of the arithmetico-geometric progression is $\frac{49}{2}$, then a_4 is equal to _____. **[JEE(Main)April,2023]**

माना a_1, a_2, 2, a_3, a_4 एक समांतर–गुणोत्तर श्रेणी है। यदि संगत गुणोत्तर श्रेणी का सार्व अनुपात 2 है तथा समांतर–गुणोत्तर श्रेणी के सभी 5 पदों का योग $\frac{49}{2}$ है, तो a_4 बराबर है _____ **[JEE(Main)April,2023]**

Ans. 16

Sol. $\frac{(a-2d)}{4}, \frac{(a-d)}{2}, a, 2(a+d), 4(a+2d)$

$a=2$

$\left(\frac{1}{4}+\frac{1}{2}+1+6\right)\times 2+(-1+2+8)d=\frac{49}{2}$

$$2\left(\frac{3}{4}+7\right)+9d=\frac{49}{2}$$

$$9d=\frac{49}{2}-\frac{62}{4}=\frac{98-62}{4}=9$$

$$d=1$$

$$\Rightarrow a_4=4(a+2d)$$

$$=16$$

39. Let $S=109+\frac{108}{5}+\frac{107}{5^2}+....+\frac{2}{5^{107}}+\frac{1}{5^{108}}$. Then the value of $\left(16S-(25)^{-54}\right)$ is equal to _____.

[JEE(Main)April,2023]

माना $S=109+\frac{108}{5}+\frac{107}{5^2}+....+\frac{2}{5^{107}}+\frac{1}{5^{108}}$ है। तो $\left(16S-(25)^{-54}\right)$ का मान बराबर है।

[JEE(Main)April,2023]

Ans. 2175

Sol. $$S=109+\frac{108}{5}+\frac{107}{5^2}+\ldots+\frac{2}{5^{107}}+\frac{1}{5^{108}}$$

$$\frac{S}{5}=\frac{109}{5}+\frac{108}{5^2}+\ldots+\ldots+\frac{2}{5^{108}}+\frac{1}{5^{109}}\cdots$$

Equation (ii) – (i) gives

$$\frac{-4S}{5}=-109+\left(\frac{1}{5}+\frac{1}{5^2}+\ldots+\frac{1}{5^{108}}+\frac{1}{5^{109}}\right)$$

$$\frac{-4S}{5}=-109+\frac{1}{5}\frac{\left(1-\left(\frac{1}{5}\right)^{109}\right)}{1-\frac{1}{5}}$$

$$\frac{4S}{5}=-109+\frac{1}{4}\left(1-\frac{1}{5^{109}}\right)$$

$$\Rightarrow 16S-(25)^{-54}=2175$$

40. For $k\in\mathbb{N}$, if the sum of the series $1+\frac{4}{k}+\frac{8}{k^2}+\frac{13}{k^3}+\frac{19}{k^4}+\ldots$ is 10, then the value of k is _____.

[JEE(Main)April,2023]

$k\in\mathbb{N}$ के लिए, यदि श्रेणी $1+\frac{4}{k}+\frac{8}{k^2}+\frac{13}{k^3}+\frac{19}{k^4}+\ldots$ का योग 10 है, तो k का मान है _____

[JEE(Main)April,2023]

Ans. 2

Sol. $10 = 1 + \frac{4}{k} + \frac{8}{k^2} + \frac{13}{k^3} + \frac{19}{k^4} + \ldots.\text{upto } \infty$

$9 = \frac{4}{k} + \frac{8}{k^2} + \frac{13}{k^3} + \frac{19}{k^4} + \ldots \text{ upto } \infty$

$\frac{9}{k} = \frac{4}{k^2} + \frac{8}{k^3} + \frac{13}{k^4} + \ldots \text{ upto } \infty$

$S = 9\left(1 - \frac{1}{k}\right) = \frac{4}{k} + \frac{4}{k^2} + \frac{5}{k^3} + \frac{6}{k^4} + \ldots\ldots\text{upto } \infty$

$\frac{S}{k} = \frac{4}{k^2} + \frac{4}{k^3} + \frac{5}{k^4} + \ldots..\text{upto } \infty$

$\left(1 - \frac{1}{k}\right)S = \frac{4}{k} + \frac{1}{k^3} + \frac{1}{k^4} + \frac{1}{k^5} + \ldots.\infty$

$9\left(1 - \frac{1}{k}\right)^2 = \frac{4}{k} + \frac{\frac{1}{k^3}}{\left(1 - \frac{1}{k}\right)}$

$9(k-1)^3 = 4k(k-1) + 1$

$k = 2$

41. The sum to 20 terms of the series $2 \cdot 2^2 - 3^2 + 2 \cdot 4^2 - 5^2 + 2 \cdot 6^2 - \ldots\ldots$ is equal to _____.

[JEE(Main)April,2023]

श्रेणी $2 \cdot 2^2 - 3^2 + 2 \cdot 4^2 - 5^2 + 2 \cdot 6^2 - \ldots\ldots$ के 20 पदों का योग है _____ **[JEE(Main)April,2023]**

Ans. 1310

Sol. $(2^2 - 3^2 + 4^2 - 5^2 + \ldots..+ 20 \text{ terms}) + (2^2 + 4^2 + \ldots..+ 10 \text{ terms})$

$= - (2 + 3 + 4 + 5 + \ldots..+21) + 4 [1 + 2^2 + \ldots..10^2]$

$= -\left[\frac{21 \times 22}{2} - 1\right] + 4 \times \frac{10 \times 11 \times 21}{6}$

$= 1 - 231 + 14 \times 11 \times 10$

$= 1540 + 1 - 231$

$= 1310$

42. If the sum of the series

$\left(\frac{1}{2} - \frac{1}{3}\right) + \left(\frac{1}{2^2} - \frac{1}{2 \cdot 3} + \frac{1}{3^2}\right) + \left(\frac{1}{2^3} - \frac{1}{2^2 \cdot 3} + \frac{1}{2 \cdot 3^2} - \frac{1}{3^3}\right) + \left(\frac{1}{2^4} - \frac{1}{2^3 \cdot 3} + \frac{1}{2^2 \cdot 3^2} - \frac{1}{2 \cdot 3^3} + \frac{1}{3^4}\right) + \ldots.. \text{ is } \frac{\alpha}{\beta}$,

where α and β are co-prime, then $\alpha + 3\beta$ is equal to _____. **[JEE(Main)April,2023]**

यदि श्रेणी

$\left(\frac{1}{2}-\frac{1}{3}\right)+\left(\frac{1}{2^2}-\frac{1}{2\cdot 3}+\frac{1}{3^2}\right)+\left(\frac{1}{2^3}-\frac{1}{2^2\cdot 3}+\frac{1}{2\cdot 3^2}-\frac{1}{3^3}\right)+\left(\frac{1}{2^4}-\frac{1}{2^3\cdot 3}+\frac{1}{2^2\cdot 3^2}-\frac{1}{2\cdot 3^3}+\frac{1}{3^4}\right)+\ldots\ldots$ का योग $\frac{\alpha}{\beta}$ है, जहाँ α तथा β सहभाज्य है, तो $\alpha + 3\beta$ का मान होगा_____ **[JEE(Main)April,2023]**

Ans. 7

Sol. $P=\left(\frac{1}{2}-\frac{1}{3}\right)+\left(\frac{1}{2^2}-\frac{1}{2.3}+\frac{1}{3^2}\right)+\left(\frac{1}{2^3}-\frac{1}{2^2.3}+\frac{1}{2.3^2}-\frac{1}{3^3}\right)+\ldots$

$P\left(\frac{1}{2}+\frac{1}{3}\right)=\left(\frac{1}{2^2}-\frac{1}{3^2}\right)+\left(\frac{1}{2^3}+\frac{1}{3^3}\right)+\left(\frac{1}{2^4}-\frac{1}{3^4}\right)+\ldots.$

$\frac{5P}{6}=\frac{\frac{1}{4}}{1-\frac{1}{2}}-\frac{\frac{1}{9}}{1+\frac{1}{3}}$

$\frac{5P}{6}=\frac{1}{2}-\frac{1}{12}=\frac{5}{12}$

$\therefore P=\frac{1}{2}=\frac{\alpha}{\beta}$ $\qquad\qquad$ $\therefore \alpha=1,\beta=2$

$\alpha+3\beta=7$

43. The number of common terms in the progressions 4,9,14,19,....., upto 25th term and 3,6,9,12,....., upto 37th term is : **[JEE(Main)Jan.2024]**

श्रेढ़ियों 4,9,14,19,....., 25 पदों तक तथा 3,6,9,12,....., 37 पदों तक उभयनिष्ठ पदों की संख्या है : **[JEE(Main)Jan.2024]**

(A) 7 (B) 9 (C) 8 (D) 5

Ans. A

Sol. 4, 9, 14, 19, ..., 124

3, 6, 9, 12,..., 111, $\left.\begin{matrix}\rightarrow CD=5\\ \rightarrow CD=3\end{matrix}\right\}$ LCM = 15

9, 24, ... ,

$9(n-1).15 \le 111$

$(n-1)15 \le 102$

$(n-1)15 = 90$

$n-1 \Rightarrow n=7$

44. The 20th term from the end of the progression $20,19\frac{1}{4},18\frac{1}{2},17\frac{3}{4},\ldots,-129\frac{1}{4}$ is : **[JEE(Main)Jan.2024]**

श्रेढ़ी $20, 19\frac{1}{4}, 18\frac{1}{2}, 17\frac{3}{4}, \ldots, -129\frac{1}{4}$ का अंत से $20^{\text{वाँ}}$ पद है : **[JEE(Main)Jan.2024]**

(A) –110 (B) –118 (C) –115 (D) –100

Ans. C

Sol. $T_1 = -129\frac{1}{4},\ d = \frac{3}{4}$

$$T_{20} = -129\frac{1}{4} + 19 \cdot \frac{3}{4}$$

$$= -\frac{-517}{4} + \frac{57}{4} = \frac{-460}{4} = -115$$

45. If in a G.P. of 64 terms, the sum of all the terms is 7 times sum of the odd terms the G.P., then the common ratio of the G.P. is equal to : **[JEE(Main)Jan.2024]**

यदि 64 पदों की एक G.P. में सभी पदों का योग, इसके विषम पदों के योग का 7 गुना है, तो G.P. का सार्व अनुपात बराबर है: **[JEE(Main)Jan.2024]**

(A) 7 (B) 6 (C) 4 (D) 5

Ans. B

Sol. $a + ar + ar^2 + ar^3 + \ldots + ar^{63}$

$$= 7\left(a + ar^2 + ar^4 \ldots + ar^{62}\right)$$

$$\Rightarrow \frac{a\left(1-r^{64}\right)}{1-r} = \frac{7a\left(1-r^{64}\right)}{1-r^2}$$

$r = 6$

46. If each term of a geometric progression $a_1, a_2, a_3, \ldots$ with $a_1 = \frac{1}{8}$ and $a_2 \neq a_1$, is the arithmetic mean of the next two terms and $S_n = a_1 + a_2 + \ldots + a_n$, then $S_{20} - S_{18}$ is equal to : **[JEE(Main)Jan.2024]**

यदि गुणोत्तर श्रेढ़ी $a_1, a_2, a_3, \ldots$ जिसमें $a_1 = \frac{1}{8}$ तथा $a_2 \neq a_1$ अगले दो पदों का समांतर माध्य है तथा $S_n = a_1 + a_2 + \ldots + a_n$ है, तो $S_{20} - S_{18}$ बराबर है : **[JEE(Main)Jan.2024]**

(A) 2^{15} (B) 2^{18} (C) -2^{15} (D) -2^{18}

Ans. C

Sol. $a_2 = \frac{a_3 + a_4}{2}$

$$\Rightarrow 2ar = ar^2 + ar^3$$

$\Rightarrow r^2 + r = 2$

$\Rightarrow r = 1 \text{ or } -2$

$\Rightarrow r \neq 1 \because (a_1 \neq a_2)$

$\Rightarrow r = -2$

Now $S_{20} - S_{18} = T_{19} + T_{20}$

$= ar^{18} + ar^{19}$

$= ar^{18}(1 + r)$

$\frac{1}{8}(-2)^{18}(-1) = -2^{15}$

47. If $\log_e a, \log_e b, \log_e c$ are in A.P. and $\log_e a - \log_e 2b, \log_e 2b - \log_e 3c, \log_e 3c - \log_e a$ are also in an A.P., then a : b : c is equal to : **[JEE(Main)Jan.2024]**

यदि $\log_e a, \log_e b, \log_e c$ एक A.P. में है तथा $\log_e a - \log_e 2b, \log_e 2b - \log_e 3c, \log_e 3c - \log_e a$ भी एक A. P. में है, तो a : b : c बराबर है : **[JEE(Main)Jan.2024]**

(A) 9 : 6 : 4 (B) 25 : 10 : 4 (C) 6 : 3 : 2 (D) 16 : 4 : 1

Ans. A

Sol. $\log_e a, \log_e b, \log_e c$ are in A.P.

$\therefore b^2 = ac$

Also $\log_e\left(\frac{a}{2b}\right), \log_e\left(\frac{2b}{3c}\right), \log_e\left(\frac{3c}{a}\right)$ are in A.P.

$\left(\frac{2b}{3c}\right)^2 = \frac{a}{2b} \times \frac{3c}{a}$

$\frac{b}{c} = \frac{3}{2}$

Putting in eq. (i) $b^2 = a \times \frac{2b}{3}$

$\frac{a}{b} = \frac{3}{2}$

$a : b : c = 9 : 6 : 4$

48. Let S_n denote the sum of first n terms of an arithemetic progression. If $S_{20} = 790$ and $S_{10} = 145$, then $S_{15} - S_5$ is : **[JEE(Main)Jan.2024]**

माना एक समान्तर श्रेढ़ी के प्रथम n पदों का योग S_n है। यदि $S_{20} = 790$ तथा $S_{10} = 145$ हैं, तो $S_{15} - S_5$ बराबर है : **[JEE(Main)Jan.2024]**

(A) 410 (B) 395 (C) 405 (D) 390

Ans. B

Sol. $S_{20} = \frac{20}{2}(2a + 19d) = 790$

$2a + 19d = 79$ _____(1)

$S_{10} = \frac{10}{2}(2a + 9d) = 145$

$2a + 9d = 29$ _____(2)

From (1) and (2) $a = -8, d = 5$

$S_{15} - S_5 = \frac{15}{2}[2a + 14d] - \frac{5}{2}[2a + 4d]$

$= \frac{15}{2}[-16 + 70] - \frac{5}{2}[-16 + 20]$

$= 405 - 10$

$= 395$

49. Let a and b be two distinct positive real numbers. Let 11th term of a GP, whose first term is a and third term is b, is equal to pth term of another GP, whose first term is a and fifth term is b. Then p is equal to :

[JEE(Main)Jan.2024]

माना a तथा b दो भिन्न धनात्मक वास्तविक संख्याऐं हैं। माना एक GP, जिसका पहला पद a तथा तीसरा पद b हैं, 11 वाँ पद, एक अन्य GP, जिसका पहला a तथा पांचवा पद b है, के p वाँ पद के बराबर है। तो p बराबर है : **[JEE(Main)Jan.2024]**

(A) 21 (B) 20 (C) 25 (D) 24

Ans. A

Sol. G.P. first : $T_1 = a$

$T_3 = b$

$\Rightarrow T_3 = ar^2 = b$

$\Rightarrow r = \pm\sqrt{\frac{b}{a}}$

G.P. second : $T_1' = a$

$T_5' = b$

$\Rightarrow T_5' = ar_1^4 = b$

$\Rightarrow r_1 = \pm\left(\frac{b}{a}\right)^{\frac{1}{4}}$

$\Rightarrow T_{11} = ar^{10}$

$T_p' = ar_1^{p-1}$

$\Rightarrow T_{11} = T_p' \Rightarrow r^{10} = r_1^{p-1}$

$\Rightarrow \frac{b^5}{a^5} = \left(\frac{b}{a}\right)^{\frac{p-1}{4}}$

$\Rightarrow \frac{p-1}{4} = 5 \qquad \Rightarrow p = 21$

50. The sum of the series $\frac{1}{1-3\cdot1^2+1^4} + \frac{2}{1-3\cdot2^2+2^4} + \frac{3}{1-3\cdot3^2+3^4} + \ldots\ldots$ upto 10-terms is :

[JEE(Main)Jan.2024]

श्रेणी $\frac{1}{1-3\cdot1^2+1^4} + \frac{2}{1-3\cdot2^2+2^4} + \frac{3}{1-3\cdot3^2+3^4} + \ldots\ldots$ 10 पदों तक का योग है : **[JEE(Main)Jan.2024]**

(A) $\frac{45}{109}$ (B) $\frac{55}{109}$ (C) $-\frac{45}{109}$ (D) $-\frac{55}{109}$

Ans. D

Sol. General term of the sequence

$T_r = \frac{r}{1-3r^2+r^4}$

$T_r = \frac{r}{r^4-2r^2+1-r^2}$

$T_r = \frac{r}{\left(r^2-1\right)^2-r^2}$

$T_r = \frac{r}{\left(r^2-r-1\right)\left(r^2+r-1\right)}$

$T_r = \frac{\frac{1}{2}\left[\left(r^2+r-1\right)-\left(r^2-r-1\right)\right]}{\left(r^2-r-1\right)\left(r^2+r-1\right)}$

$= \frac{1}{2}\left[\frac{1}{r^2-r-1} - \frac{1}{r^2+r-1}\right]$

Sum of 10 terms,

$\sum_{r=1}^{10} T_r = \frac{1}{2}\left[\frac{1}{-1} - \frac{1}{109}\right] = \frac{-55}{109}$

51. For $0 < c < b < a$, let $(a+b-2c)x^2 + (b+c-2a)x + (c+a-2b) = 0$ and $\alpha \neq 1$ be one of its root. Then, among the two statements :

(I) If $\alpha \in (-1, 0)$, then b cannot be the geometric mean of a and c.

(II) If $\alpha \in (0, 1)$, then b may be the geometric mean of a and c. **[JEE(Main)Jan.2024]**

(A) Neither (I) nor (II) is true. (B) Only (I) is true

(C) Only (II) is true. (D) Both (I) and (II) are true

$0 < c < b < a$, के लिए माना $(a + b - 2c)x^2 + (b + c - 2a)x + (c + a - 2b) = 0$ का एक मूल $\alpha \neq 1$ है। तो दो दो कथनों : **[JEE(Main)Jan.2024]**

(I) यदि $\alpha \in (-1, 0)$ है, तो a तथा b का गुणोत्तर माध्य b नहीं हो सकता।

(II) यदि $\alpha \in (0, 1)$ है, तो a तथा c का गुणोत्तर माध्य b हो सकता है।

(A) (I) तथा (II) दोनों सत्य हैं (B) केवल (I) सत्य है

(C) केवल (II) सत्य है (D) न तो (I) न ही (II) सत्य हैं

Ans. D

Sol. $(a + b - 2c)x^2 + (b + c - 2a)x + (c + a - 2b) = 0$. Hence $0 < c < b < a$

When $x = 1$

$(a + b - 2c) + (b + c - 2a) + (c + a - 2b) = 0$

So one root of equation is $\beta = 1$ and other root is given $\alpha \neq 1$

$$\alpha\beta = \frac{c+a-2b}{a+b-2c}$$

$$\therefore \alpha = \frac{c+a-2b}{a+b-2c}$$

when $-1 < \alpha < 0$

$$-1 < \frac{c+a-2b}{a+b-2c} < 0 \qquad (\because\ c < a,\ c < b,\ 2c < a + b)$$

$-a - b + 2c < c + a - 2b < 0$

$2b > a + c$

$$b > \frac{a+c}{2} \qquad \left(G = \sqrt{ac}\right)$$

$(G < A)$

So b can not be GM of a and c.

When $0 < \alpha < 1$

$$0 < \frac{c+a-2b}{a+b-2c} < 1$$

$0 < c + a - 2b < a + b - 2c$

$2b < c + a$

$$b < \frac{a+c}{2};\ b = \sqrt{ac}\ ; \qquad GM < AM$$

So b may be GM of a and c.

52. Let 2nd, 8th and 44th terms of a non-constant A. P. be respectively the 1st, 2nd and 3rd terms of a G.P. If the first term

of the A.P. is 1, then the sum of its first 20 terms is equal to : **[JEE(Main)Jan.2024]**

माना एक भिन्न पदों की A. P. का दूसरा, आठवाँ तथा चवालिसवाँ पद, एक G. P. के क्रमशः पहला, दूसरा तथा तीसरा पद हैं। यदि A. P. का प्रथम पद 1 है, तो इसके प्रथम 20 पदों का योग है : **[JEE(Main)Jan.2024]**

(A) 990 (B) 980 (C) 960 (D) 970

Ans. D

Sol. $a + d, a + 7d, a + 43d$ are $1^{st}, 2^{nd}, 3^{rd}$ term of a G.P.

$$\frac{a+7d}{a+d} = \frac{a+43d}{a+7d}$$

$\Rightarrow (a + 7d)^2 = (a + 43d)(a + d)$

$\Rightarrow a^2 + 49d^2 + 14ad = a^2 + 44ad + 43d^2$

$\Rightarrow 6d^2 = 30ad$

$\Rightarrow d = 0, 5$

But $d \neq 0$

$\Rightarrow d = 5, a = 1$

$$S_{20} = \frac{20}{2}\left[2 + (19)^5\right]$$

$= 10(95 + 2) = 970$

53. Let 3, a, b, c be in A.P. and 3, a – 1, b + 1, c + 9 be in G.P. Then, the arithmetic mean of a, b and c is : **[JEE(Main)Jan.2024]**

माना 3, a, b, c एक A.P. में हैं तथा 3, a – 1, b + 1, c + 9 एक G.P. में हैं। तो a, b तथा c का समांतर माध्य है : **[JEE(Main)Jan.2024]**

(A) –4 (B) 11 (C) 13 (D) –1

Ans. B

Sol. 3, a, b, c be in A.P.

$a – 3 = b – a = c – b$

3, a – 1, b + 1, c + 9 in G.P.

$$\frac{a-1}{3} = \frac{b+1}{a-1} = \frac{c+9}{b+1}\left(a \neq 1, b \neq -1\right)$$

$(a – 1)^2 = 3 (b + 1)$

$a^2 + 1 – 2a = 3b + 3$

$a^2 – 8a + 7 = 0$

$(a – 1)(a – 7) = 0$

$a = 7$ as $a \neq 1$

$b = 2a – 3$

$= 14 – 3$

= 11

c = 2b – a = 22 – 7 = 15

Mean of a, b, c

$$\frac{a+b+c}{3}=\frac{7+11+15}{3}=\frac{33}{3}=11$$

54. Let α and β be the roots of the equation $px^2 + qx - r = 0$, where $p \neq 0$. If p, q and r be the consecutive terms of a non constant G.P. and $\frac{1}{\alpha}+\frac{1}{\beta}=\frac{3}{4}$ then the value of $(\alpha-\beta)^2$ is : **[JEE(Main)Jan.2024]**

माना समीकरण $px^2 + qx - r = 0$, $p \neq 0$ के मूल α तथा β हैं। यदि p, q तथा r एक परिवर्तनीय (non constant) G.P. के क्रमागत पदप हैं तथा $\frac{1}{\alpha}+\frac{1}{\beta}=\frac{3}{4}$ है, तो $(\alpha-\beta)^2$ का मान हैं : **[JEE(Main)Jan.2024]**

(A) 9 (B) 8 (C) $\frac{80}{9}$ (D) $\frac{20}{3}$

Ans. C

Sol. $px^2 + qx - r = 0$

roots are α, β

$$\frac{1}{\alpha}+\frac{1}{\beta}=\frac{3}{4}$$

$$4(\alpha+\beta)=3\alpha\beta$$

$$4\left(\frac{-q}{p}\right)=3\left(\frac{-r}{p}\right)$$

$4q = 3r$

$(p, q, r) \equiv (A, AR, AR^2)$

$4AR = 3AR^2$

$$\Rightarrow R=\frac{4}{3}$$

$\Rightarrow$ quadratic equation will be

$$Ax^2+\frac{4A}{3}x-\frac{16}{9}A=0$$

$9x^2 + 12x - 16 = 0$

$$(\alpha-\beta)^2=(\alpha+\beta)^2-4\alpha\beta$$

$$=\left(-\frac{12}{9}\right)^2-4\left(-\frac{16}{9}\right)=\frac{80}{9}$$

55. Let S_n denote the sum of the first n terms of an arithmetic progression. If $S_{10} = 390$ and the ratio of the tenth and the fifth terms is 15 : 7, then $S_{15} - S_5$ is equal to : **[JEE(Main)Jan.2024]**

माना एक समान्तर श्रेढ़ी के प्रथम n पदों का योग S_n है। यदि $S_{10} = 390$ तथा दसवें और पाँचवें पदों का अनुपात 15 : 7 है, तो $S_{15} - S_5$ बराबर है : **[JEE(Main)Jan.2024]**

(A) 690 (B) 890 (C) 790 (D) 800

Ans. C

Sol. $S_{10} = 390$

$$\frac{a_{10}}{a_5} = \frac{15}{7}$$

$$\frac{a+9d}{a+4d} = \frac{15}{7}$$

$$7a + 63d = 15a + 60d$$

$$8a - 3d = 0 \quad \text{_____(1)}$$

Also $S_{10} = 390$

$5[2a + 9d] = 390$

$2a + 9d = 78$ _____(2)

From equation (1) & (2)

$d = 8$

$\Rightarrow a = 3$

$$\text{So, } S_{15} - S_5 = \frac{15}{2}[2a + 14d] - \frac{5}{2}[2a + 4d]$$

$= 5[2a + 19d]$

Putting values of a and d in above equation

$S_{15} - S_5 = 5[2(3) + 19(8)]$

$S_{15} - S_5 = 790$

56. If $8 = 3 + \frac{1}{4}(3+p) + \frac{1}{4^2}(3+2p) + \frac{1}{4^3}(3+3p) + ...\infty$, then the value of p is _____.

[JEE(Main)Jan.2024]

यदि $8 = 3 + \frac{1}{4}(3+p) + \frac{1}{4^2}(3+2p) + \frac{1}{4^3}(3+3p) + ...\infty$ है, तो p का मान है _____

[JEE(Main)Jan.2024]

Ans. 9

Sol. $$8 = \frac{3}{1-\frac{1}{4}} + \frac{p.\frac{1}{4}}{\left(1-\frac{1}{4}\right)^2}$$

(Sum of infinite terms of A.G.P. $= \frac{a}{1-r} + \frac{dr}{(1-r)^2}$)

$\Rightarrow \frac{4p}{9} = 4 \qquad \Rightarrow p = 9$

57. Let $\alpha = 1^2 + 4^2 + 8^2 + 13^2 + 19^2 + 26^2 + ...$upto 10 terms and $\beta = \sum_{n=1}^{10} n^4$. If $4\alpha - \beta = 55k + 40$, then k is equal to _____. **[JEE(Main)Jan.2024]**

माना $\alpha = 1^2 + 4^2 + 8^2 + 13^2 + 19^2 + 26^2 + ...$ 10 पदों तक तथा $\beta = \sum_{n=1}^{10} n^4$ हैं। यदि $4\alpha - \beta = 55k + 40$ है, तो k बराबर है _____ **[JEE(Main)Jan.2024]**

Ans. 353

Sol. $\alpha = 1^2 + 4^2 + 8^2 ...$

$t_n = an^2 + bn + c$

$1 = a + b + c$

$4 = 4a + 2b + c$

$8 = 9a + 3b + c$

On solving we get, $a = \frac{1}{2}, b = \frac{3}{2}, c = -1$

$$\alpha = \sum_{n=1}^{10} \left(\frac{n^2}{2} + \frac{3n}{2} - 1 \right)^2$$

$$4\alpha = \sum_{n=1}^{10} \left(n^2 + 3n - 2 \right)^2 \ , \ \beta = \sum_{n=1}^{10} n^4$$

$$4\alpha - \beta = \sum_{n=1}^{10} \left(6n^3 + 5n^2 - 12n + 4 \right) = 55(353) + 40$$

58. Let S_n be the sum to n-terms of an arithmetic progression 3, 7, 11, ... If $40 < \left(\frac{6}{n(n+1)} \sum_{k=1}^{n} S_k \right) < 42$, then n equals _____. **[JEE(Main)Jan.2024]**

माना समान्तर श्रेढी 3, 7, 11, ... के प्रथम n पदों का योग S_n है। यदि $40 < \left(\frac{6}{n(n+1)} \sum_{k=1}^{n} S_k \right) < 42$ है, तो n बराबर है _____ **[JEE(Main)Jan.2024]**

Ans. 9

Sol. $S_k = 3 + 7 + 11 + ...$ up to k terms

$= \frac{k}{2}\left[6 + (k-1)4 \right] = k(2k+1)$

$$\sum_{k=1}^{n}\left(2k^2+k\right)=\frac{2n(n+1)(2n+1)}{6}+\frac{n(n+1)}{2}$$

$40 < 2(2n + 1) + 3 < 42$

$35 < 4n < 37$

$8.75 < n < 9.25$

$n = 9$

59. Let a, b, c be the lengths of three sides of a triangle satisfying the condition $\left(a^2+b^2\right)x^2-2b(a+c)x+\left(b^2+c^2\right)=0$. If the set of all possible values of x is the interval (α, β), then $12(\alpha^2+\beta^2)$ is equal to _____. **[JEE(Main)Jan.2024]**

माना एक त्रिभुज की तीन भुजाओं की लम्बाईयाँ a, b, c हैं, जो $\left(a^2+b^2\right)x^2-2b(a+c)x+\left(b^2+c^2\right)=0$ के संतुष्ट करती हैं। यदि x के सभी संभव मानों का समुच्चय, अंतराल (α, β) है, तो $12(\alpha^2+\beta^2)$ बराबर है _____

[JEE(Main)Jan.2024]

Ans. 36

Sol. $\left(a^2+b^2\right)x^2-2b(a+c)x+\left(b^2+c^2\right)=0$

$\Rightarrow \left(a^2x^2-2ab+b^2\right)+\left(b^2x^2-2bcx+c^2\right)=0$

$\Rightarrow (ax-b)^2+(bx-c)^2=0$

$\Rightarrow x=\frac{b}{a}$ and $x=\frac{c}{b}\Rightarrow b=ax$

$\Rightarrow \frac{b}{a}=\frac{c}{b}\Rightarrow b^2=ac$

Using triangular inequality

$a + b > c, b + c > a, c + a > b$

$\Rightarrow a + ax > bx$

$\Rightarrow a + ax > ax^2 > 0$

$\Rightarrow x^2 - x - 1 < 0$

Similarly, $ax + bx > a$

$\Rightarrow x^2 + x - 1 > 0$

$c + a > b$

$\Rightarrow x^2 - x + 1 > 0$ (always true)

$\Rightarrow x^2 - x - 1 < 0$ and $x^2 + x - 1 > 0$

$\Rightarrow x\in\left(\frac{1-\sqrt{5}}{2},\frac{1+\sqrt{5}}{2}\right)$ and $x\in\left(\frac{\sqrt{5}-1}{2},\frac{\sqrt{5}+1}{2}\right)$

$\Rightarrow x\in\left(\frac{\sqrt{5}-1}{2},\frac{\sqrt{5}+1}{2}\right)$

$\Rightarrow \alpha^2+\beta^2=\left(\frac{\sqrt{5}-1}{2}\right)^2+\left(\frac{\sqrt{5}+1}{2}\right)^2=3$

$\Rightarrow 12(\alpha^2+\beta^2)=36$

$\Rightarrow x^2+x-1>0$

$c+a>b$

$\Rightarrow x^2-x+1>0$ (always true)

$\Rightarrow x^2-x-1<0$ and $x^2+x-1>0$

$\Rightarrow x\in\left(\frac{1-\sqrt{5}}{2},\frac{1+\sqrt{5}}{2}\right)$ and $x\in\left(\frac{\sqrt{5}-1}{2},\frac{\sqrt{5}+1}{2}\right)$

$\Rightarrow x\in\left(\frac{\sqrt{5}-1}{2},\frac{\sqrt{5}+1}{2}\right)$

$\Rightarrow \alpha^2+\beta^2=\left(\frac{\sqrt{5}-1}{2}\right)^2+\left(\frac{\sqrt{5}+1}{2}\right)^2=3$

$\Rightarrow 12(\alpha^2+\beta^2)=36$

60. Let 3,7,11,15,....,403 and 2,5,8,11,....,404 be two arithmetic progressions. Then the sum, of the common terms in them, is equal to _____. **[JEE(Main)Jan.2024]**

माना 3, 7, 11, 15,....,403 तथा 2, 5, 8, 11,...., 404 दो समांतर श्रेढि.याँ हैं। तो इनमें उभयनिष्ठ पदों का योग है _____

[JEE(Main)Jan.2024]

Ans. 6699

Sol. 3,7,11,15,....403

2,5,8,11,....404

So common term AP

11,23,35,....,395

$\Rightarrow 395=11+(n-1)12$

$\Rightarrow n=33$

$\text{Sum}=\frac{33}{2}\left[2\times11+(32)12\right]$

$=\frac{33}{2}\left[22+384\right]$

$=6699$

61. If three successive terms of a G.P. with common ratio $r(r>1)$ are the lengths of the sides of a triangle and [r] denotes the greatest integer less than or equal to r, then $3[r]+[-r]$ is equal to _____.

[JEE(Main)Jan.2024]

यदि सार्व अनुपात r(r > 1) की एक G. P. के तीन क्रमागत पद एक त्रिभुज की भुजाओं की लबाइयाँ हैं तथा [r] महत्तम पूर्णांक ≤ r है, तो 3[r] + [–r] बराबर है _____ **[JEE(Main)Jan.2024]**

Ans. A

Sol. Let three terms of GP are $\frac{a}{r}$, a, ar(r > 1)

Sum of two smaller sides > third side

$\Rightarrow \frac{a}{r} + a > ar$

$\Rightarrow 1 + r > r^2$

$\Rightarrow r^2 - r - 1 < 0$

$\Rightarrow \frac{1-\sqrt{5}}{2} < r < \frac{1+\sqrt{5}}{2}$

but $r > 1 \Rightarrow r \in \left(1, \frac{1+\sqrt{5}}{2}\right)$

$\Rightarrow$ [r] = 1 and [–r] = –2

So 3[r] + [– r] = 3 – 2 = 1

62. Let the first three terms 2, p and q with $q \neq 2$, of a G.P. be respectively the 7th, 8th and 13th terms of an A.P. If the 5th term of the G.P. is the nth term of the A.P., then n is equal to : **[JEE(Main)April,2024]**

माना एक G. P. के प्रथम तीन पद 2, p तथा q, जहाँ $q \neq 2$, एक A.P. के क्रमशः 7वाँ, 8वाँ तथा 13वाँ पद हैं। यदि G.P. का 5वाँ पद, A.P. का nवाँ पद है, तो n बराबर है : **[JEE(Main)April,2024]**

(A) 163 (B) 169 (C) 151 (D) 177

Ans. A

Sol. 2, p, q are in GP $\Rightarrow p^2 = 2q$ _____(1)

Let first term and common difference of AP are a and d respectively

$\Rightarrow$ $t_7 = a + 6d = 2$ _____(2)

$t_8 = a + 7d = p$ _____(3)

$t_{13} = a + 12d = q$ _____(4)

by (2) & (3) d = (p –2)

and by(2) & (4) 6d = (q – 2)

$\Rightarrow$ 6(p –2) = q –2

$\Rightarrow$ 6p = q +10 _____(5)

by (1) and (5) we have p = 2 or p = 10

but if p = 2 then q = 2 (rejected)

when p = 10 we have q = 50

so GP is 2, 10, 50 , ...

also by (2) and (3) we have

d= 8 and a = –46

Now 5th term of GP = n^{th} term of AP

$\Rightarrow 2(5)^4 = -46 + (n-1)\,8$

$\Rightarrow \frac{1250+46}{8}+1 = n$

$\Rightarrow n = 163$

63. The value of $\frac{1\times 2^2 + 2\times 3^2 + ... + 100\times (101)^2}{1^2\times 2 + 2^2\times 3 + ... + 100^2\times 101}$ is : **[JEE(Main)April,2024]**

$\frac{1\times 2^2 + 2\times 3^2 + ... + 100\times (101)^2}{1^2\times 2 + 2^2\times 3 + ... + 100^2\times 101}$ का मान है : **[JEE(Main)April,2024]**

(A) $\frac{31}{30}$ (B) $\frac{306}{305}$ (C) $\frac{32}{31}$ (D) $\frac{305}{301}$

Ans. D

Sol. $\frac{1\times 2^2 + 2\times 3^2 + \ldots + 100\times (101)^2}{1^2\times 2 + 2^2\times 3 + \ldots + 100^2\times 101} = \frac{\sum_{r=1}^{100} r(r+1)^2}{\sum_{r=1}^{100} r^2(r+1)}$

$= \frac{\sum_{r=1}^{100}\left(r^3+2r^2+r\right)}{\sum_{r=1}^{100}\left(r^3+r^2\right)} \quad = \frac{\left(\frac{n(n+1)}{2}\right)^2 + \frac{2.n(n+1)(2n+1)}{6} + \frac{n(n+1)}{2}}{\left(\frac{n(n+1)}{2}\right)^2 + \frac{n(n+1)(2n+1)}{6}}$

$= \frac{\frac{n(n+1)}{2}\left[\frac{n(n+1)}{2} + \frac{2}{3}\cdot(2n+1)+1\right]}{\frac{n(n+1)}{2}\left[\frac{n(n+1)}{2} + \frac{(2n+1)}{3}\right]}$; Put n = 100

$$= \frac{\frac{100(101)}{2} + \frac{2}{3}(201) + 1}{\frac{100 \times 101}{2} + \frac{201}{3}} = \frac{5185}{5117} = \frac{305}{301}$$

64. Let three real numbers a, b, c be in arithmetic progression and a + 1, b, c + 3 be in geometric progression. If a > 10 and the arithmetic mean of a, b and c is 8, then the cube of the geometric mean of a, b and c is :

[JEE(Main)April,2024]

माना तीन वास्तविक संख्याएँ a, b, c समांतर श्रेणी में हैं तथा a + 1, b, c + 3 गुणोत्तर श्रेणी में है यदि a > 10 है तथा a, b तथा c का समांतर माध्य 8 है, तो a, b तथा c के गुणोत्तर माध्य का घन है : **[JEE(Main)April,2024]**

(A) 316 (B) 312 (C) 128 (D) 120

Ans. D

Sol. $2b = a + c$ _____(1)

$b^2 = (a + 1)(c + 3)$ _____(2)

$\frac{a+b+c}{3} = 8$ _____(3)

$b = 8$

$\Rightarrow$ $ac + 3a + c + 3 = 64$

$3a + c + ac = 61$ _____(4)

$a + c = 16$

$c = 16 - a$

from equation (4)

$3a + 16 - a + a(16 - a) = 61$

$\Rightarrow$ $(a - 15)(a - 3) = 0$

$a = 15$ $(a > 10)$

$\Rightarrow$ $a = 15, b = 8, c = 1$

$$\left((a \cdot b \cdot c)^{\frac{1}{3}}\right)^3 = 15 \times 8 \times 1 = 120$$

65. If $\frac{1}{\sqrt{1}+\sqrt{2}} + \frac{1}{\sqrt{2}+\sqrt{3}} + ... + \frac{1}{\sqrt{99}+\sqrt{100}} = m$ and $\frac{1}{1.2} + \frac{1}{2.3} + ... + \frac{1}{99.100} = n$, then then point (m, n) lies on the line :

[JEE(Main)April,2024]

यदि $\frac{1}{\sqrt{1}+\sqrt{2}}+\frac{1}{\sqrt{2}+\sqrt{3}}+.....+\frac{1}{\sqrt{99}+\sqrt{100}}=m$ तथा $\frac{1}{1.2}+\frac{1}{2.3}+.....+\frac{1}{99.100}=n$ है, तो बिन्दु (m, n) किस रेखा पर है : **[JEE(Main)April,2024]**

(A) $11(x-2)-100(y-1)=0$ (B) $11(x-1)-100y=0$

(C) $11x-100y=0$ (D) $11(x-1)-100(y-2)=0$

Ans. C

Sol. $m=\left(\sqrt{2}-1\right)+\left(\sqrt{3}-\sqrt{2}\right)+\left(\sqrt{4}-\sqrt{3}\right)+.....\left(\sqrt{100}-\sqrt{99}\right)$

$m=10-1=9$

$n=\frac{1}{1.2}+\frac{1}{2.3}+\frac{1}{3.4}+.....+\frac{1}{99.100}$

$=\left(1-\frac{1}{2}\right)+\left(\frac{1}{2}-\frac{1}{3}\right)+\left(\frac{1}{3}-\frac{1}{4}\right)+.....+\left(\frac{1}{99}-\frac{1}{100}\right)$

$=1-\frac{1}{100}=\frac{99}{100}$

66. For $x\geq 0$, the least value of K, for which $4^{1+x}+4^{1-x},\frac{K}{2},16^{x}+16^{-x}$ are three consecutive terms of an A.P., is equal to : **[JEE(Main)April,2024]**

K का न्यूनतम मान, जिसके लिए $4^{1+x}+4^{1-x},\frac{K}{2},16^{x}+16^{-x}$ (जहाँ $x\geq 0$) एक A.P. के तीन क्रमागत पद हैं, है : **[JEE(Main)April,2024]**

(A) 10 (B) 4 (C) 16 (D) 8

Ans. A

Sol. $2b=a+c\Rightarrow k=4\left(4^{x}+\frac{1}{4^{x}}\right)+\left(16^{x}+\frac{1}{16^{x}}\right)$

A.M $\geq$ G.M

$K\geq 4(2)+(2)=10$

Least value of $k=10$

67. Let ABC be an equilateral triangle. A new triangle is formed by joining the middle points of all sides of the triangle ABC and the same process is repeated infinitely many times. If P is the sum of perimeters and Q is be the sum of areas of all the triangles formed in this process, then : **[JEE(Main)April,2024]**

माना ABC एक समबाहु त्रिभुज है। त्रिभुज ABC की सभी भुजाओं के मध्य बिन्दुओं को जोड़ने से एक नया त्रिभुज बनाया गया है तथा

इसी प्रक्रिया को अनन्त बार दोहराया गय है। यदि इस प्रक्रिया से बने सभी त्रिभुजों के परिमापों का योग P है तथा क्षेत्रफलों का योग Q है, तो : **[JEE(Main)April,2024]**

(A) $P = 36\sqrt{3}Q^2$ (B) $P^2 = 36\sqrt{3}Q$ (C) $P^2 = 72\sqrt{3}Q$ (D) $P^2 = 6\sqrt{3}Q$

Ans. B

Sol. Area of first equilateral $\Delta = \frac{\sqrt{3}}{4}a^2$

Area of Second equilateral $\Delta = \frac{\sqrt{3}}{4}\left(\frac{a}{2}\right)^2 = \frac{\sqrt{3}a^2}{16}$

Area of Second equilateral

Sum of area $= \frac{\sqrt{3}}{4}a^2 + \frac{\sqrt{3}}{16}a^2 + \frac{\sqrt{3}}{64}a^2 + ...$

$= \frac{\sqrt{3}}{4}\left(1 + \frac{1}{4} + \frac{1}{16} + ...\right)$

$= \frac{\sqrt{3}}{4}\left(\frac{1}{1-\frac{1}{4}}\right)$

$Q = \frac{\sqrt{3}}{4}a^2\left(\frac{1}{1-\frac{1}{4}}\right)$

$Q = \frac{\sqrt{3}}{4}a^2\left(\frac{4}{3}\right)$

$Q = \frac{a^2}{\sqrt{3}}$ .....(1)

Perimeter of first D 3a $D = \frac{a}{2} + \frac{a}{2} + \frac{a}{2} = \frac{3a}{2}$

Perimeter of second $D = \frac{a}{2} + \frac{a}{2} + \frac{a}{2} = \frac{3a}{2}$

Perimeter of third $D = \frac{a}{4} + \frac{a}{4} + \frac{a}{4} = \frac{3a}{4}$

Sum of perimeter $P = P = 3a + \frac{3a}{2} + \frac{3a}{4} + ...$

$$P = 3a\left(1+\frac{1}{2}+\frac{1}{4}+.....\right)$$

$$P = \frac{3a}{1-\frac{1}{2}} = 6a$$

$P^2 = 36a^2$ ....(2)

From (1) & (2)

$$\frac{Q}{P^2}\frac{\alpha^2}{\sqrt{3}.36\alpha^2}$$

$$P^2 = 36\sqrt{3}Q$$

68. A software company sets up m number of computer systems to finish an assignment in 17 days. If 4 computer systems crashed on the start of the second day, 4 more computer systems crashed on the start of the third day and so on, then it took 8 more days to finish the assignment. The value of m is equal to :

[JEE(Main)April,2024]

एक कार्य को 17 दिन में समाप्त करने के लिए एक सॉफ्टवेयर कंपनी ने m कंप्यूटर स्थापित किए। यदि दूसरे दिन के शुरू में ही 4 कंप्यूटर खराब हो गए, तीसरे दिन के शरू में 4 और कंप्यूटर खराब हो गए, और इसी प्रकार से आगे के दिनों में भी, तो कार्य को समाप्त करने में 8 दिन अधिक लगे। m का मान बराबर है :

[JEE(Main)April,2024]

(A) 150 (B) 125 (C) 180 (D) 160

Ans. A

Sol. $17m = m + (m-4) + (m-4\times 2)$ ....+... $(m-4\times 24)$

$$8m = \frac{4.24.25}{2} = 150$$

69. In an increasing geometric progression of positive terms, the sum of the second and sixth terms is $\frac{70}{3}$ and the product of the third and fifth terms is 49. Then the sum of the 4th, 6th and 8th terms is equal to :

[JEE(Main)April,2024]

धनात्मक पदों की एक वर्धमान गुणोत्तर श्रेढ़ी में, दूसरे तथा छठे पदों का योग $\frac{70}{3}$ है तथा तीसरे और पाँचवें पदों का गुणनफल 49 है। तो चौथे, छठे तथा आठवें पदों का योग है :

[JEE(Main)April,2024]

(A) 91 (B) 96 (C) 78 (D) 84

Ans. A

Sol. $T_2 + T_6 = \frac{70}{3}$

$ar + ar^5 = \frac{70}{3} \quad \Rightarrow ar\left(1+r^4\right) = \frac{70}{3}$ ...(1)

$T_3 \times T_5 = 49$

$ar^2 \cdot ar^4 = 49 \quad \Rightarrow (ar)^2 r^4 = 49$ ...(2)

from (1) and (2) $\frac{\left(1+r^4\right)^2}{r^4} = \frac{70\times 70}{3\times 3\times 49}$

$\left(r^2 + \frac{1}{r^2}\right) = \frac{10}{3} = 3 + \frac{1}{3}$ $\qquad r^2 = 3,\ r = +\sqrt{3},\ a = \frac{7}{3\sqrt{3}}$

$T_4 + T_6 + T_8 = a(r^3 + r^5 + r^7) = \frac{7}{3\sqrt{3}}\left(3\sqrt{3} + 9\sqrt{3} + 27\sqrt{3}\right) = 91$

70. If the sum of the series $\frac{1}{1.(1+d)} + \frac{1}{(1+d)(1+2d)} + \ldots\ldots + \frac{1}{(1+9d)(1+10d)}$ is to 5, then 50d is equal to :

[JEE(Main)April,2024]

यदि श्रेणी $\frac{1}{1.(1+d)} + \frac{1}{(1+d)(1+2d)} + \ldots\ldots + \frac{1}{(1+9d)(1+10d)}$ का योग 5 है, तो 50d बराबर है :

[JEE(Main)April,2024]

(A) 10 (B) 5 (C) 15 (D) 20

Ans. B

Sol. $S = \frac{1}{d}\sum_{r=1}^{10} \frac{d}{(1+(r-1)d)(1+rd)} = 5$

$\frac{1}{d}\sum_{r=1}^{10} \frac{(1+rd)-(1+(r-1)d)}{(1+(r-1)d)(1+rd)} = 5$

$\frac{1}{d}\left[\sum_{r=1}^{10}\left[\frac{1}{(1+(r-1)d)} - \frac{1}{1+rd}\right]\right] = 5$

$\frac{1}{1} - \frac{1}{1+d} + \frac{1}{1+d} - \frac{1}{1+2d} + \ldots\ldots + \frac{1}{1+9d} - \frac{1}{1+10d} = 5d$

$1 - \frac{1}{1+10d} = 5d$

$10d = 1$

$50d = 5$

71. Let a, ar, ar^2, ... be an infinite G.P. If $\sum_{n=0}^{\infty} ar^n = 57$ and $\sum_{n=0}^{\infty} a^3r^{3n} = 9747$, then a + 18r is equal to :

[JEE(Main)April,2024]

माना a, ar, ar^2, ... एक अनंत गुणोत्तर श्रेणी है। यदि $\sum_{n=0}^{\infty} ar^n = 57$ तथा $\sum_{n=0}^{\infty} a^3r^{3n} = 9747$ हैं, तो a + 18r बराबर है:

[JEE(Main)April,2024]

(A) 46 (B) 31 (C) 38 (D) 27

Ans. B

Sol. $\sum_{n=0}^{\infty} a^n = 57$

$a + ar + ar^2 + ...\infty \text{ terms} = 57$

$\frac{a}{1-r} = 57$ _____(I)

$\sum_{n=0}^{\infty} a^3r^{3n} = 9747$

$a^3 \cdot a^3 \cdot r^3 + a^3 \cdot r^6 + ...\infty \text{ terms} = 9746$

$\frac{a^3}{1-r^3} = 9746$ _____(II)

$\frac{(I)^3}{(II)} \Rightarrow \frac{a^3}{\frac{(1-r)^3}{a^3}} = \frac{57^3}{9717} = 19$

On solving, $r = \frac{2}{3}$ and $r = \frac{3}{2}$ (rejected)

$a = 19$

$\therefore a + 18r = 19 + 18 \times \frac{2}{3} = 31$

72. The $a_1, a_2, a_3, ...$ be in the arithemetic progression of positive terms.

Let $A_k = a_1^2 - a_2^2 + a_3^2 - a_4^2 + ... + a_{2k-1}^2 - a_{2k}^2$.

If $A_3 = -153, A_5 = -435$ and $a_1^2 + a_2^2 + a_3^2 = 66$, then $a_{17} - A_7$ is equal to_____.

[JEE(Main)April,2024]

माना $a_1, a_2, a_3, ...$ एक समांतर श्रेणी है, जिसके पद धनात्मक हैं।

माना $A_k = a_1^2 - a_2^2 + a_3^2 - a_4^2 + ... + a_{2k-1}^2 - a_{2k}^2$ है। यदि $A_3 = -153$, $A_5 = -435$ तथा $a_1^2 + a_2^2 + a_3^2 = 66$ हैं, तो $a_{17} - A_7$ बराबर है _____ **[JEE(Main)April,2024]**

Ans. 910

Sol. $d \rightarrow$ common diff.

$A_k = -kd[2a + (2k - 1)d]$

$A_3 = -153$

$\Rightarrow 153 = 13d[2a + 5d]$

$51 = d[2a + 5d]$ ...(1)

$A_5 = -435$

$435 = 5d[2a + 9d]$

$87 = d[2a + 9d]$

(B) – (1)

$36 = 4d^2$

$d = 3, a = 1$

$a_{17} - A_7 = 49 - [-7.3[2 + 39]] = 910$

73. Let the first term of a series be $T_1 = 6$ and its rth term $T_r = 3T_{r-1} + 6^r$, $r = 2, 3, ..., n$. If the sum of the first n terms of this series is $\frac{1}{5}\left(n^2 - 12n + 39\right)\left(4 \cdot 6^n - 5 \cdot 3^n + 1\right)$, then n is equal to _____.

[JEE(Main)April,2024]

माना किसी श्रेणी का प्रथम पद $T_1 = 6$ तथा उसका r-वाँ पद $T_r = 3T_{r-1} + 6^r$, $r = 2, 3, ..., n$ है। यदि इस श्रेणी के प्रथम n पदों का योग $\frac{1}{5}\left(n^2 - 12n + 39\right)\left(4 \cdot 6^n - 5 \cdot 3^n + 1\right)$ है, तो n बराबर है _____ **[JEE(Main)April,2024]**

Ans. 6

Sol. $T_1 = 6$

$T_2 = 3 \times 6 + 6^2 = 6^2 \times \left(\frac{1}{2} + 1\right) = \frac{3}{2} \times 6^2$

Similarly $T_3 = 6^3 \times \frac{7}{4}$

Now $S_n = \left(\frac{2-1}{2^0}\right)6 + \left(\frac{2^2 - 2}{2}\right)6^2 + \left(\frac{2^3 - 1}{2^2}\right)6^3 + ...$

$$S_n = \left(2\times 6 + 2\times 6^2 + 2\times 6^3 + \ldots..\text{n terms}\right) - \left(\frac{6}{2^0} + \frac{6^2}{2^1} + \frac{6^3}{2^2} + \ldots\ldots.. \text{ n terms}\right)$$

$$= \frac{12}{5}\times 6^n - \frac{12}{5} - \frac{6}{2}\times 3^n + 3 = \frac{3}{5}\left[4\times 6^n - 5\left(3^n\right) + 1\right]$$

But $S_n = \frac{1}{5}\left(n^2 - 12n + 39\right)\left(4.6^n - 5.3^n + 1\right)$

So, $n^2 - 12n + 39 = 3$

$(n - 6)^2 = 0$

$n = 6$

74. If $S(x) = (1 + x) + 2(1 + x)^2 + 3(1 + x)^3 + \ldots\ldots + 60\ (1 + x)^{60}$, $x \neq 0$, and $(60)^2\ S(60) = a(b)^b + b$, where $a, b \in N$, then $(a + b)$ equal to _____. **[JEE(Main)April,2024]**

यदि $S(x) = (1 + x) + 2(1 + x)^2 + 3(1 + x)^3 + \ldots\ldots + 60\ (1 + x)^{60}$, $x \neq 0$ तथा $(60)^2\ S(60) = a(b)^b + b$, जहाँ $a, b \in N$, तो $(a + b)$ बराबर है _____ **[JEE(Main)April,2024]**

Ans. 3660

Sol. $S(x) = (1 + x) + 2(1 + x)^2 + 3(1 + x)^3 + \ldots\ldots + 60\ (1 + x)^{60}$

$(1 + x)S(x) = 1(1 + x)^2 + (1 + x)^3 + \ldots.. + 59\ (1 + x)^{60} + 60(1 + x)^{61}$

– – – – – –

$S(x)\ [1 - 1 - 1] = [(1 + x) + (1 + x)^2 + \ldots\ldots + (1 + x)^{60}] - 60\ (1 + x)^{61}$

$$(-x)\left(S(x)\right) = \frac{(1+x)\left[(1+x)^{60} - 1\right]}{(1+x-1)} - 60(1+x)^{61}$$

$$(-x)\left(S(x)\right) = \frac{(1+x)\left[(1+x)^{60} - 1\right]}{x} - 60(1+x)^{61}$$

$$xS(x) = 60(1+x)^{61} - \frac{(1+x)\left[(1+x)^{60} - 1\right]}{x}$$

Multiplying x on both side,

$x^2S(x) = 60x\ (1 + x)^{61} - (1 + x)\ [(1 + x)^{60} - 1]$

Putting $g = 60$

$(60)^2S(60) = 60 \times 60(61)^{61} - (61)[61^{60} - 1]$

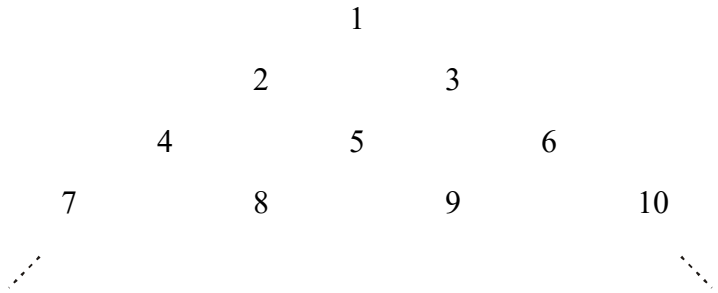
$= 60 \times 60(61)^{61} - (61) \cdot 61^{60} + 61$

$= (61)^{61}[60 \times 60 - 1] + 61$

$= (3600 - 1) \cdot 61^{61} + 61$

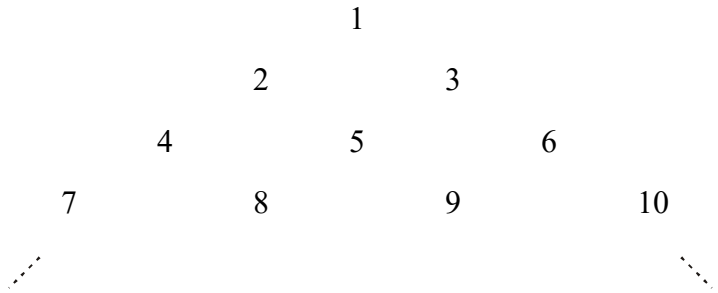
$a = 3600 - t,\ b = 61 \Rightarrow a + b = 3660$

75. Let the positive integers be written in the form



If the kth row contains exactly k numbers for every natural number k, then the row in which the number 5310 will be, is _____. **[JEE(Main)April,2024]**

माना धनात्मक पूर्णांक इस प्रकार लिखे गये है

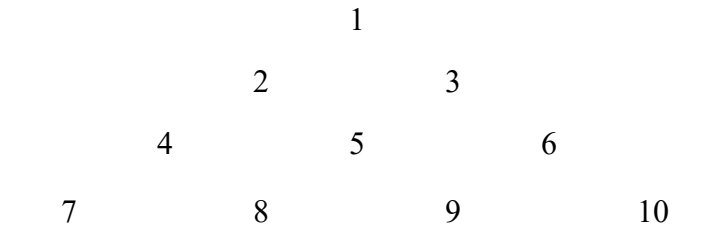


प्रत्येक प्राकृत संख्या k के लिए kवीं पंक्ति ठीक k संख्याऐं समाहित करती है, तब संख्या 5310 किस पंक्ति में होगी _____

[JEE(Main)April,2024]

Ans. 103

Sol.



$S = 1 + 3 + 6 + 10, \ldots\ldots\ldots\ldots, T_n$

$S = 1 + 3 + 6 + \ldots\ldots + T_{n-1} + T_n$

$0 = 1 + 2 + 3 + 4 + \ldots\ldots\ldots + n - T_n$

$$T_n = \frac{n(n+1)}{2}$$

for $n = 102$

$$T_n = \frac{102(103)}{2} \Rightarrow 5253$$

for n = 103

$$T_n = \frac{103 \times 104}{2} = 5336$$

K = 103

76. An arithmetic progression is written in the following way

2

5 8

11 14 17

20 23 26 29

– –

– –

The sum of all the terms of the 10th row is _____. **[JEE(Main)April,2024]**

एक समांतर श्रेढ़ी को निम्न प्रकार से लिखा जाता है

2

5 8

11 14 17

20 23 26 29

– –

– –

तो दसवीं पंक्ति के सभी पदों का योग है _____. **[JEE(Main)April,2024]**

Ans. 1505

Sol. First term is each row form pattern

2, 5, 11, 20

3 6 9 $\longrightarrow$ forms an AP

$\Rightarrow T_n = an^2 + bn + c$

$\Rightarrow T_1 = a + b + c = 2$

$\Rightarrow T_2 = 4a + 2b + c = 5$

$\Rightarrow T_3 = 9a + 3b + c = 11$

$\Rightarrow 3a + b = 3$

$5a + b = 6$

$\Rightarrow 2a = 3 \Rightarrow a = \frac{3}{2},\ b = \frac{-3}{2} \Rightarrow c = 2$

$\Rightarrow T_n = \frac{3}{2}n^2 - \frac{3(n)}{2} + 2 \Rightarrow \frac{3n^2 - 3n + 4}{2}$

$T_{10} = \frac{3\times100 - 3\times10 + 4}{2} = \frac{274}{2} = 137$

Terms in 10th row is 10 with 3 differences

$\Rightarrow$ 137, 140, 143 ...

$\Rightarrow S_{10} = \frac{10}{2}\left(2\times137 + (10-1)\times3\right)$

$= 5(274 + 27) = 5 \times 301 = 1505$

77. If $\left(\frac{1}{\alpha+1} + \frac{1}{\alpha+2} + \ldots\ldots + \frac{1}{\alpha+1012}\right) - \left(\frac{1}{2\cdot1} + \frac{1}{4\cdot3} + \frac{1}{6\cdot5} + \ldots\ldots + \frac{1}{2024\cdot2023}\right) = \frac{1}{2024}$, then α is equal to _____. **[JEE(Main)April,2024]**

यदि $\left(\frac{1}{\alpha+1} + \frac{1}{\alpha+2} + \ldots\ldots + \frac{1}{\alpha+1012}\right) - \left(\frac{1}{2\cdot1} + \frac{1}{4\cdot3} + \frac{1}{6\cdot5} + \ldots\ldots + \frac{1}{2024\cdot2023}\right) = \frac{1}{2024}$ है, तो α बराबर है _____. **[JEE(Main)April,2024]**

Ans. 1011

Sol. $\left(\frac{1}{\alpha+1} + \frac{1}{\alpha+2} + \ldots\ldots + \frac{1}{\alpha+1012}\right) - \left(\frac{1}{2\cdot1} + \frac{1}{4\cdot3} + \frac{1}{6\cdot5} + \ldots\ldots + \frac{1}{2024\cdot2023}\right) = \frac{1}{2024}$

$\downarrow$

Let it be z

Considering,

$Z = \frac{1}{2.1} + \frac{1}{4.3} + \frac{1}{6.5} + \ldots \frac{1}{2024.2023}$

$Z = 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \frac{1}{5} - \frac{1}{6} + \ldots \frac{1}{2023} - \frac{1}{2024}$

$Z = \left(1 + \frac{1}{3} + \frac{1}{5} + \ldots\ldots \frac{1}{2023}\right) - \left(\frac{1}{2} + \frac{1}{4} + \frac{1}{6} + \ldots \frac{1}{2024}\right)$

$$Z=\left(1+\frac{1}{3}+\frac{1}{5}+\ldots\frac{1}{2023}\right)-\frac{1}{2}\left(1+\frac{1}{2}+\frac{1}{3}+\ldots\frac{1}{2012}\right)$$

Adding and subtracting $\left(\frac{1}{2}+\frac{1}{4}+\ldots\frac{1}{2022}\right)$

$$Z=\left(1+\frac{1}{2}+\frac{1}{3}+\frac{1}{4}+\ldots\frac{1}{2023}\right)-\frac{1}{2}\left(1+\frac{1}{2}+\frac{1}{3}+\ldots\frac{1}{2012}\right)-\frac{1}{2}\left(1+\frac{1}{2}+\frac{1}{3}+\ldots\frac{1}{2011}\right)$$

$$Z=\left(1+\frac{1}{2}+\frac{1}{3}+\ldots\frac{1}{2023}\right)-\left(1+\frac{1}{2}+\frac{1}{3}+\ldots\frac{1}{2011}\right)-\frac{1}{2024}$$

$$Z=\frac{1}{1012}+\frac{1}{1013}\cdots\frac{1}{2023}-\frac{1}{2024}$$

Now, put in equation (1) $\left(\frac{1}{\alpha+1}+\frac{1}{\alpha+2}+\ldots\frac{1}{\alpha+10/2}\right)-\left(\frac{1}{1012}+\frac{1}{1013}+\ldots.\frac{1}{2023}\right)+\frac{1}{2024}$

Now, comparing above equation with equation (1)

$\alpha = 1011$